

The Macroeconomic Impact of Paying in Dollars in Emerging Economies

Johar Arrieta Vidal*

University of Michigan

March 9, 2026

Abstract

In partially dollarized emerging economies, households earn in local currency but save in dollars while major durables—housing and vehicles—are also dollar-priced. This denomination appears welfare-improving: dollar savings and durable replacement costs appreciate with the exchange rate, providing a natural hedge. I develop and estimate a quantitative heterogeneous-agent model with lumpy durable adjustment and portfolio choice using Uruguayan microdata. Switching to local-currency pricing or restricting dollar access would, in my model, result in meaningfully sized welfare losses.

Keywords: Dollar prices, Durable goods, Lumpy Adjustment.

JEL Classification: C63, E21, E22, F31, G51

*I thank John Leahy, Dmitriy Stolyarov, Stephen Terry, Rudi Bachmann, Lucia Casal, Rafael Guntin, Eva Janssens, Valeria Morales, Pablo Ottonello, Matthew Schwartzman, Toni Whited, Wei Xiang, and seminar participants at the University of Michigan and the PhD-Economics Virtual Seminar for their helpful comments. I am also grateful to Zuleika Ferre for facilitating access to the Uruguayan Household Financial Survey (*Encuesta Financiera de los Hogares Uruguayos*, EFHU) and to the Rackham Graduate Student Research Grant for financial support. All errors are my own. E-mail: joharav@umich.edu

1 Introduction

Why do households in low-inflation emerging economies continue to save and transact around the dollar, even after domestic inflation has stabilized? A central reason is that currency denomination still shapes household risk exposure: many routine flows remain in local currency, while major purchases and part of savings remain tied to the dollar. I study how that denomination structure transmits exchange-rate risk to household spending, portfolios, and welfare.

This pattern persists long after stabilization in countries like Uruguay, Bolivia, and Peru, suggesting that the dollar serves not only as a store of value but also as a hedge against currency risk (Calvo and Vegh, 1992; Levy Yeyati and Ize, 2005; Drenik and Perez, 2021). Durable goods sit at the center of that hedge: they are simultaneously consumption goods and the largest balance-sheet items for most households, linking the real and financial sides of the household portfolio.

When the peso depreciates, the cost of replacing housing or a vehicle rises in local-currency terms, while dollar savings appreciate by the same factor—providing insurance when it is most needed. The balance-sheet logic becomes especially salient during large depreciation episodes, such as Uruguay’s 2002 crisis, when exchange-rate movements generate sharp real wealth revaluations (Drenik et al., 2018). Holding dollars, however, carries a proportional convenience cost for routine peso-denominated transactions (Cravino and Levchenko, 2017; Castellares and Toma, 2020). This friction generates the data’s distinctive pattern—limited dollar participation, but high dollarization among participants—and disciplines identification of the portfolio wedge.

I develop a quantitative household model with three ingredients: (i) lumpy durable adjustment with partial irreversibility, (ii) sticky dollar pricing of durables with exchange-rate pass-through, and (iii) portfolio choice between peso and dollar assets subject to transaction frictions in a small-open-economy framework. In the quantitative implementation, durables refer to housing and vehicles. Together, these ingredients generate state-dependent saving and adjustment behavior.

Dollar pricing operates through three linked channels. It creates a portfolio hedging motive—dollar assets appreciate precisely when durable replacement becomes more expensive—while also generating state-dependent precaution, as households accumulate dollar buffers ahead of anticipated adjustment. At the same time, exchange-rate movements directly shift the local-currency cost of durable purchases, amplifying adjustment frictions by widening inaction regions and making replacement more infrequent and lumpy.

Four empirical facts discipline the quantitative analysis: dollar participation is limited but intensive dollarization is high; major durables are predominantly dollar-priced even in a low-inflation

environment; dollar holdings rise with household income; and durable adjustment is infrequent, with long tenure spells. The model is designed to replicate each directly. The portfolio wedge generates limited participation with high intensive dollarization, dollar pricing matches the denomination of durable purchases in the data, income risk produces heterogeneous hedging demand across households, and nonconvex adjustment costs generate lumpy durable replacement.

I estimate key parameters by simulated method of moments (SMM) using microdata from the Uruguayan Household Financial Survey (*Encuesta Financiera de los Hogares Uruguayos*, EFHU), which uniquely records liquid assets by currency denomination. Then, I compute generalized impulse response functions (GIRFs) and use counterfactual experiments to isolate the welfare effects of dollar pricing and dollar access. The GIRFs show that depreciations trigger an immediate balance-sheet revaluation, leading households to defer durable adjustment, accumulate liquid buffers, and temporarily reduce their dollar share as valuation effects dominate on impact. These responses then unwind as the exchange rate mean-reverts and households gradually return toward the inaction region. Thus, the GIRFs provide the model-based analogue of the household adjustment pressures that become salient during large depreciation episodes, including crisis environments such as Uruguay in 2002.

The counterfactuals indicate that dollar pricing of durables is *welfare-improving*. Since durable prices and dollar savings co-move with the exchange rate, switching to local-currency pricing would break that hedge and generate a welfare loss of about 11%. Restricting dollar access would generate a separate loss of about 9%, while higher adjustment frictions would imply a loss above 20%. These magnitudes are economically large: they are in the range of, or larger than, standard gains from trade in developing economies (Arkolakis et al., 2012).

The results imply a policy trade-off. Restricting foreign-currency deposits may strengthen systemic financial stability, but it also increases household exposure to exchange-rate risk by removing the key hedging instrument. For Uruguay, a small open economy in which world interest rates and the exchange rate are plausibly exogenous, partial equilibrium is the natural baseline. As a robustness check, I solve a general-equilibrium version in the spirit of Aiyagari (1994), closing the domestic peso asset market while keeping dollar savings and durable purchases on the world market. Endogenizing the domestic rate and wage attenuates partial-equilibrium welfare magnitudes; the GE effects range from 0.06% (near Lucas (1987) welfare benchmark) to 0.54% (similar to incomplete-markets business-cycle costs (Imrohoroglu, 1989)), remaining economically meaningful. Across both PE and GE, the welfare impacts of dollar pricing and dollar access are sizable and robust.

The paper contributes to three strands of research. First, it extends the literature on lumpy durable adjustment and household portfolio choice by introducing currency denomination as a new friction

(Caballero, 1993; Berger and Vavra, 2015; Kaplan et al., 2020; McKay and Wieland, 2021; Baley and Blanco, 2021; Beraja and Wolf, 2021). Existing work documents how nonconvex adjustment costs generate infrequent, large durable purchases and shape aggregate consumption dynamics, but abstracts from the currency of denomination. I link the adjustment directly to portfolio dollarization, generating empirically testable cross-sectional predictions.

Second, the paper contributes to the quantitative literature on dollarization and currency pricing (Calvo and Vegh, 1992; Levy Yeyati and Ize, 2005; Drenik and Perez, 2021, 2024; Castellares and Toma, 2020; Salomao and Varela, 2022). Prior papers document dollar pricing is pervasive even in low-inflation economies and that foreign-currency holdings respond to exchange-rate risk, but typically treat portfolio choice and durable investment as separate. I connect them by modeling how dollar-priced durables generate endogenous demand for dollar savings. This also complements the household balance-sheet evidence in Drenik et al. (2018), which shows that large exchange-rate movements redistribute real resources across households in Uruguay.

Third, the paper is related to quantitative heterogeneous-agent models of small open economies. Zhou (2022) studies how exchange-rate movements redistribute wealth across Uruguayan households, using balance-sheet estimates from De Rosa (2019). My approach complements theirs by emphasizing the portfolio and durable-adjustment margins that determine household risk exposure.

Section 2 describes the data and empirical facts that motivate the model. Section 3 defines the model. Section 4 covers estimation, fit, and GIRFs. Section 5 introduces the counterfactuals, including the peso-pricing experiment. Section 6 presents robustness checks, extensions, and discussion, including a general-equilibrium approach. Section 7 concludes.

2 Data

This section presents the data and empirical facts that motivate the model. Four regularities stand out: limited dollar participation with high intensive dollarization, pervasive dollar pricing of major durables, a positive relationship between income and dollar holdings, and infrequent durable adjustment with long tenure spells. Section 3 formalizes the mechanisms behind these patterns.

I construct the empirical moments using the Uruguayan Household Financial Survey (*Encuesta Financiera de los Hogares Uruguayos*, EFHU). It was conducted in three waves between 2012 and 2017 by the *Universidad de la República*. It is nationally representative, covers roughly 20,000 households, and complements the household survey by collecting balance-sheet information.

A distinctive feature of EFHU is that it records liquid asset holdings separately in pesos and U.S. dollars, together with detailed information on debts, durable ownership, values, and the timing of durable purchases. The survey design allows measurement of both the level and the currency composition of household balance sheets, which is central for analyzing household dollarization.

From the survey I construct measures of: (i) liquid savings in pesos and dollars and the implied dollar share of liquid assets; (ii) total household income; and (iii) durable ownership, values, and tenure (years since last upgrade). I harmonize variables across survey waves to ensure consistency. I express all monetary values in U.S. dollars using official exchange rates at the time of each survey wave. Most moments used in the quantitative analysis are ratios (e.g. dollar shares, durable shares, adjustment frequencies) and are therefore invariant to the unit of account.

EFHU consists of repeated cross-sections rather than a true household panel. Accordingly, I compute empirical moments at the cross-sectional level within each wave and pool them using survey weights. All statistics reported in the section use household expansion factors.

To benchmark the survey against macroeconomic aggregates, I complement EFHU with official statistics from the Central Bank of Uruguay on deposit dollarization, inflation, interest rates, and both real and nominal exchange rates. I also use expenditure-flow data for durables and non-durables from the 2016 *Encuesta de Gastos e Ingresos de los Hogares (ENGIH)*. To pin down the average durable price, I rely on external pricing evidence from Drenik and Perez (2021).

2.1 Empirical facts

I document four empirical facts that motivate the model and discipline the quantitative analysis.

Fact 1: Dollar participation is limited, but intensive dollarization is high. In EFHU Wave 2, the participation margin is 10.6%, while the intensive margin among participants is high: 75.4%. The pattern reconciles micro and macro facts. According to the Central Bank of Uruguay (BCU), roughly one-half to two-thirds of *aggregate* bank deposits are denominated in foreign currency—a concentration driven by households that do hold dollars and hold them at high shares. At the household level, the unconditional mean dollar share is 8.0%. Figure 1 shows both margins: a large mass at zero in the full sample and a high conditional distribution among participants, with substantial mass near one. The bimodal pattern is consistent with threshold behavior driven by proportional transaction costs, as formalized below in Section 3.

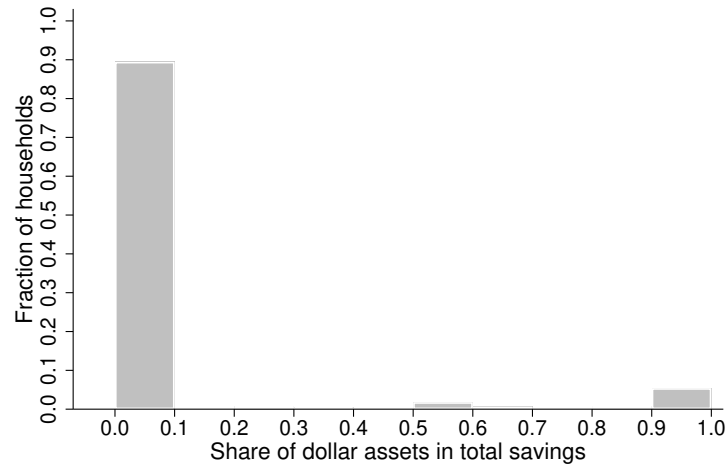


Figure 1: Unconditional distribution of portfolio dollarization

Notes: Wave 2 (EFHU 2013), survey-weighted. Unconditional histogram of the household dollar share of liquid assets (dollar deposits / total liquid assets); y-axis is the fraction of households. A large mass at zero reflects non-participation ($\approx 89\%$ of households hold no dollar deposits), while among participants the intensive margin is high (conditional mean $\approx 75\%$). $N = 22,058$ households with positive liquid assets.

Fact 2: Dollar pricing of durables. Durable goods are predominantly quoted in dollars, while non-durables remain peso-denominated. Survey responses on durable values and complementary price data show vehicles, appliances, and housing-related expenditures are largely priced in dollars (Drenik and Perez, 2021). Table 1 places Uruguay in a cross-country perspective: despite low inflation, it combines high deposit dollarization with pervasive dollar pricing of durables, indicating that dollar pricing is a persistent structural feature rather than a purely inflationary phenomenon.

Table 1: Cross-country pricing and deposit dollarization

Country	Non-durables	Vehicles	Real estate	Inflation (%)	Dollar deposits (%)
Argentina	0	9	76	47.6	32
Bolivia	42	67	83	1.5	—
Paraguay	60	57	33	3.2	45
Peru	4	68	54	2.2	40
Uruguay	35	85	88	7.6	75

Notes: Entries report the share (in percent) of listed prices denominated in U.S. dollars at the largest retailer in South America. Non-durables correspond to grocery and consumer goods prices, vehicles correspond to used-car listings; real estate corresponds to residential property listings in capital cities (Drenik and Perez, 2021). Inflation is CPI inflation in 2018 (World Bank, 2024). Dollar deposits denote the share of foreign-currency deposits in total bank deposits in 2020, from national central bank statistics. A dash indicates unavailable data.

Fact 3: Heterogeneity in dollarization across households. Dollar participation and portfolio shares vary systematically across the income distribution. Dollar saving is not confined to high-income households: middle- and lower-income households also hold dollar assets, though typically in smaller amounts. Figure 2 shows an increasing pattern. The heterogeneity motivates modeling income risk and endogenous portfolio choice.

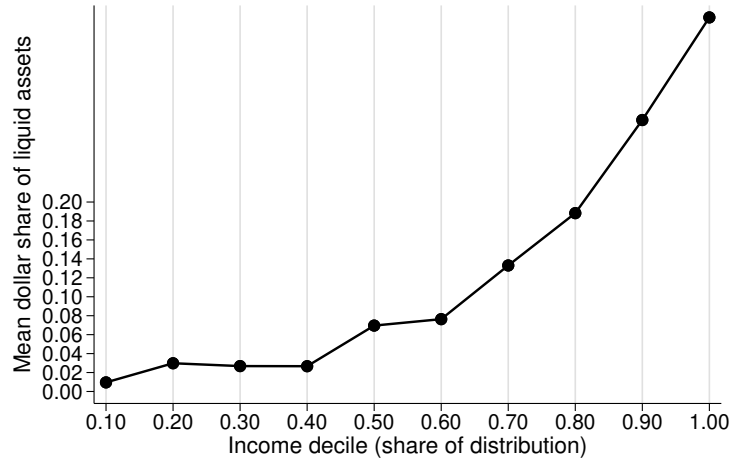


Figure 2: Dollar share of liquid assets by household income decile

Notes: Each point reports the mean dollar share of liquid assets for an income decile. Income deciles are built from total household labor income within each EFHU wave (2012, 2014, 2017) and then pooled. The upward slope shows that higher-income households hold larger dollar shares, consistent with fixed participation costs or stronger hedging demand. Sample: (N=22,058) households with positive liquid assets, using survey expansion weights.

Fact 4: Infrequent durable adjustment. Durable adjustment is infrequent. Most households report last upgrading their main durable more than a decade ago. Figure 3 shows the distribution of durable tenure (years since last update). Since durables are priced in dollars, exchange-rate movements raise the local-currency cost of replacement and widen inaction regions.

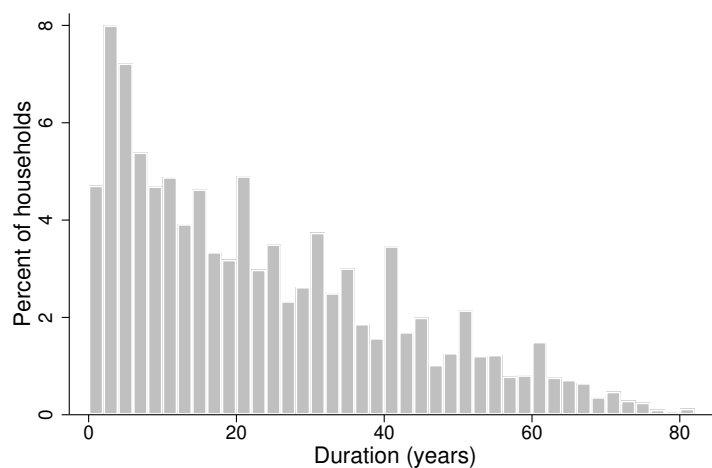


Figure 3: Durable updating is infrequent

Notes: Histogram of years since last durable upgrade, computed from self-reported tenure in EFHU (2012–2017). The median tenure exceeds 20 years, indicating infrequent adjustment. Durables include housing and vehicles. Survey-weighted; sample restricted to households with positive income and non-negative durable holdings. $N = 22,058$.

Table 2 reports key descriptive statistics from EFHU. Durables constitute the bulk of household wealth, while dollar deposits account for a substantial share of liquid savings.

Table 2: Descriptive statistics from EFHU (2012–2017), survey-weighted

Statistics (survey-weighted)	Value	Unit
Dollar deposit participation	0.106	ratio
Dollar deposit share, conditional mean	0.754	ratio
Dollar deposit share, unconditional mean	0.08	ratio
Durable value / wealth (mean)	0.83	ratio
Durable tenure (average years)	23.2	years
Housing value (median)	\$150,000	USD
Vehicle value (median)	\$8,000	USD
Quarterly income (median)	\$4,922	USD

Notes: Statistics use EFHU survey weights. Dollar moments come from Wave 2 (EFHU 2013), where the currency-share variable is continuous and comparable to the model. The dollar deposit share equals foreign-currency deposits over total liquid financial assets; participation is the share of households with positive dollar deposits; the conditional mean is computed among participants; and the unconditional mean is computed across all households. Durables include housing and vehicles at self-reported market values. Wealth equals liquid assets plus durables, income is total quarterly household income, and all monetary values are converted to U.S. dollars.

Taken together, the facts show dollarization in Uruguay is simultaneously a portfolio phenomenon, a pricing phenomenon, and a heterogeneous one. The combination of detailed microdata, stable inflation, and salient exchange-rate risk provides a sharp empirical environment to identify the interaction between dollar pricing, portfolio choice, and lumpy durable adjustment. The empirical regularities motivate the following model and quantitative exercises.

3 Model

I develop a partial-equilibrium household model with incomplete markets, lumpy durable adjustment, idiosyncratic earnings risk, and exchange-rate shocks in a small-open-economy framework. The model is designed to match the four facts documented in Section 2: limited but intensive dollarization, dollar pricing of durables, income-gradient dollar holdings, and infrequent lumpy durable adjustment. The environment builds on Kaplan and Violante (2014); Berger and Vavra (2015); Kaplan et al. (2020) and McKay and Wieland (2021), and introduces two features motivated by South American data.

First, I model durable goods as priced in U.S. dollars with complete exchange-rate pass-through. In the Uruguayan context, durables correspond to vehicles and housing. Vehicles are imported and uniformly dollar-priced. Housing is traded with prices quoted in dollars even though construction services are local; the empirical evidence in Drenik and Perez (2024) confirms 80–90% of durable

listings in Uruguay are dollar-denominated and that dollar-priced items tend to remain longer on the market. The model treats the composite durable as dollar-priced, which is consistent with the dollar-pricing facts for both categories in the Uruguayan data.

Second, households can hedge exchange-rate risk by holding liquid wealth in dollars, but dollar liquidity is costly or inconvenient for routine non-durable purchases (Kehoe and Nicolini, 2021). The model makes dollarization endogenous through a portfolio choice over currency composition.

A key modeling choice is tractability with endogenous dollarization. Rather than treating peso and dollar deposits as separate dynamic states, I summarize the liquid portfolio by total liquid wealth in local-currency units and model currency composition as a within-period choice. The approach keeps the hedging motive but avoids adding two asset states.

3.1 Household maximization problem

Households have CRRA utility over non-durable consumption $c_t \geq 0$ and their stock of durables $d_t \geq 0$:

$$u(c_t, d_t) = \frac{(c_t^\gamma d_t^{1-\gamma})^{1-\gamma}}{1-\gamma}, \quad (3.1)$$

where $\gamma \in (0, 1)$ controls the expenditure share on non-durables and $\gamma > 0$ is the coefficient of relative risk aversion.

Log earnings follow an AR(1):

$$\log y_t = \rho_y \log y_{t-1} + \varepsilon_t^y, \quad \varepsilon_t^y \sim \mathcal{N}(0, \sigma_y^2). \quad (3.2)$$

The nominal exchange rate e_t (local currency per USD) follows

$$\log e_t = \rho_e \log e_{t-1} + \varepsilon_t^e, \quad \varepsilon_t^e \sim \mathcal{N}(0, \sigma_e^2). \quad (3.3)$$

The durable has a sticky USD price p_d , so its local-currency price is $P_t^d \equiv e_t p_d$. The non-durable good is the numeraire.¹

¹Durables refer to housing and vehicles, the two largest balance-sheet categories in EFHU. Vehicles are traded internationally at dollar prices; housing is non-tradable but is conventionally quoted and settled in U.S. dollars in Uruguay, a monetary practice rooted in the country's currency history (Drenik and Perez, 2021). Dollar denomination of a non-tradable good does not require goods-market arbitrage: sellers of large, long-lived assets prefer a stable unit of account. The model's mechanism is the same for both: $P_t^d = e_t p_d$ rises with the exchange rate regardless of tradability.

3.1.1 Reduced-state liquid wealth and portfolio choice

Let w_t denote *total liquid wealth in local-currency units* at the beginning of period t ,

$$w_t \equiv a_t + e_t a_t^{\$},$$

where a_t are peso deposits and $a_t^{\$}$ are USD deposits.

Within the period, households choose a dollar share $s_t \in [0, 1]$ for their *end-of-period* savings decision. Let $w'_t \geq 0$ denote savings carried from t to $t+1$ (before returns are realized). The household allocates w'_t across currencies:

$$a'_t = (1 - s_t) w'_t, \quad a_t^{\$} = \frac{s_t w'_t}{e_t}.$$

Holding USD positions entails a proportional wedge $\kappa \in (0, 1)$. It captures ongoing costs of maintaining dollar liquidity—such as account fees, limited merchant acceptance, or the inconvenience of currency conversion for routine purchases. Since κ is proportional to dollar holdings, households with small dollar positions face low costs and may prefer corner solutions at $s = 0$ or $s = 1$, generating the bimodal portfolio distribution observed in the data.

The portfolio choice determines next period's liquid wealth:

$$w_{t+1} = (1 + r) (1 - s_t) w'_t + (1 + r^{\$}) s_t w'_t \frac{e_{t+1}}{e_t} - \kappa s_t w'_t. \quad (3.4)$$

The formulation makes dollarization endogenous while keeping (w_t, d_{t-1}, e_t, y_t) as the dynamic state. Finally, I impose a no-borrowing constraint $w'_t \geq 0$ to isolate the portfolio hedging channel from debt considerations.

3.1.2 Durables: depreciation and adjustment

Durables deliver utility services and depreciate over time. If the household does not adjust, the durable stock evolves according to

$$d_t = (1 - \delta) d_{t-1}, \quad (3.5)$$

where $\delta \in (0, 1)$ is the baseline depreciation rate.

If the household adjusts, it chooses a new durable stock $d^* \geq 0$ and liquidates the depreciated stock at a resale value subject to partial irreversibility. Specifically, selling the depreciated stock yields

a fraction $(1 - f)$ of its replacement value, with $f \in (0, 1)$ interpreted as a resale haircut, which generates inaction regions and lumpy adjustment (Caballero and Engel, 1993).

3.1.3 Recursive formulation

The state is

$$S = (w, d_{-1}, e, y).$$

Each period the household chooses non-durable consumption c , end-of-period savings w' , a portfolio share s , and (if adjusting) a new durable stock d .

The value function is

$$V(S) = \max \left\{ V^{\text{adjust}}(S), V^{\text{noadjust}}(S) \right\}. \quad (3.6)$$

If the household adjusts, it chooses (c, w', d, s) :

$$V^{\text{adjust}}(S) = \max_{c, w', d, s \in [0, 1]} \left\{ u(c, d) + \beta \mathbb{E}[V(S') | S] \right\}, \quad (3.7)$$

$$\text{s.t.} \quad c + w' + P^d d = y + (1 + r)w + (1 - f)P^d(1 - \delta)d_{-1}, \quad (3.8)$$

$$w^+ = (1 + r)(1 - s)w' + (1 + r^s)s w' \frac{e'}{e} - \kappa s w', \quad (3.9)$$

$$c \geq 0, \quad w' \geq 0, \quad d \geq 0.$$

where $P^d \equiv e p_d$ and w^+ denotes next-period liquid wealth.

If the household does not adjust, durables follow (3.5) and the household chooses (c, w', s) :

$$V^{\text{noadjust}}(S) = \max_{c, w', s \in [0, 1]} \left\{ u(c, (1 - \delta)d_{-1}) + \beta \mathbb{E}[V(S') | S] \right\}, \quad (3.10)$$

$$\text{s.t.} \quad c + w' = y + (1 + r)w, \quad (3.11)$$

$$w^+ = (1 + r)(1 - s)w' + (1 + r^s)s w' \frac{e'}{e} - \kappa s w', \quad (3.12)$$

$$c \geq 0, \quad w' \geq 0.$$

Earnings and the exchange rate evolve according to (3.2)–(3.3). Expectations are taken over (e', y') conditional on (e, y) .

3.2 Main mechanisms

The model features four interconnected mechanisms that link dollar pricing, portfolio choice, and durable adjustment. This subsection follows the order of the four empirical facts in Section 2: limited but intensive dollarization, dollar pricing of durables, heterogeneity in dollar holdings across households, and infrequent durable adjustment. Taken together, these mechanisms determine how exchange-rate shocks pass through to household spending, portfolio rebalancing, and the timing of durable replacement.

Mechanism 1: Endogenous dollarization as hedging. Households can partially insure against currency risk by holding liquid wealth in dollars. When the peso depreciates, dollar-denominated savings appreciate in local-currency terms, providing resources precisely when durable replacement becomes expensive. The portfolio share s_t balances the hedging benefit against the proportional wedge κ , which captures the reduced transactional usefulness of dollar assets for everyday peso-denominated consumption. In equilibrium, households tilt toward dollars when exchange-rate volatility is high, but the wedge prevents full dollarization. This mechanism is the model counterpart to Fact 1: it rationalizes a household distribution with many non-participants, yet high dollar shares among those who do hold dollars.

Mechanism 2: Dollar pricing and exchange-rate pass-through. I model durable goods as priced in U.S. dollars with complete pass-through to local-currency prices: $P_t = e_t p_d$. The assumption reflects the pervasive dollar pricing of durables in Uruguay documented by Drenik and Perez (2021, 2024). Vehicles are internationally traded and directly dollar-priced; housing is non-tradable but conventionally denominated in dollars—a monetary pricing convention rather than a tradability requirement, as discussed in the pricing footnote above and in Appendix G. In both cases the local-currency adjustment cost is $P_t^d = e_t p_d$, so exchange-rate movements pass through one-for-one to the peso cost of durable replacement, making this the model counterpart to Fact 2 in the data. A depreciation (rise in e_t) raises the peso cost of replacing durables, effectively tightening the household’s budget constraint when it chooses to adjust. It creates an asymmetry: depreciations make adjustment more costly, while appreciations make it cheaper.

Importantly, the hedging relationship implies dollar pricing of durables creates a natural hedge. When the durable price and the value of dollar savings both co-move with the exchange rate, dollar holdings insure households against the very risk that dollar pricing creates. The complementarity between dollar assets and dollar-priced durables turns out to have first-order welfare implications, which I quantify in Section 5.

Mechanism 3: Portfolio-durable interaction. The preceding mechanisms are not independent—they interact through the household’s wealth accumulation. Liquid assets provide self-insurance against both income shocks and exchange-rate risk, while durables provide utility services but are costly to adjust and partially illiquid. The wealth transition equation (3.4) ties portfolio composition to the timing of durable adjustment: households who anticipate adjusting soon have stronger incentives to hold dollars (to hedge replacement costs), while households deep in the inaction region may prefer peso liquidity for consumption smoothing.

The interaction generates state-dependent portfolio behavior that varies with the household’s position in the durable gap distribution. It also sharpens shock transmission: after a depreciation, households close to the adjustment boundary are the ones most likely to defer replacement and rebuild buffers. In cross section, that state dependence helps rationalize Fact 3: higher-income and higher-wealth households optimally choose larger dollar positions because they are better able to finance both precautionary buffers and durable replacement.

Mechanism 4: Lumpy adjustment through irreversibility. Durable adjustment is infrequent and lumpy due to frictions. Partial irreversibility—modeled as a resale haircut f —means households selling durables recover only a fraction of replacement value. Partial irreversibility generates asymmetric adjustment: households are reluctant to downsize even when their durable stock exceeds the target. Hence, there are inaction regions in which households tolerate deviations from their optimal durable stock rather than paying adjustment costs. Households delay replacement until the deviation becomes sufficiently large, then make discrete, lumpy changes (Caballero, 1993). This is the direct model counterpart to Fact 4, the long tenure spells observed in the survey.

3.3 Policy functions

The mechanisms above manifest in the model’s policy functions. Since the state space is four-dimensional, I present conditional slices in the (w, e) plane to illustrate the main predictions.

Figure 4 displays the portfolio dollarization decision $s^*(w, e)$ in (w, e) space. Three zones are distinguished: all-peso ($s^* = 0$), all-dollar ($s^* = 1$), and interior ($s^* \in (0, 1)$). The figure reveals a predominantly binary optimal portfolio choice, consistent with the mass points at zero and one observed in the EFHU data (Figure 1). At low wealth levels households tend to hold pesos, while at medium-to-high wealth the all-dollar solution dominates across most exchange-rate levels. The interior region is mostly concentrated near the low-wealth boundary. The pattern arises endogenously: nonlinear hedging demand generated by dollar-priced durables pushes households

to corners even under a proportional transaction cost that would otherwise admit smooth interior solutions.

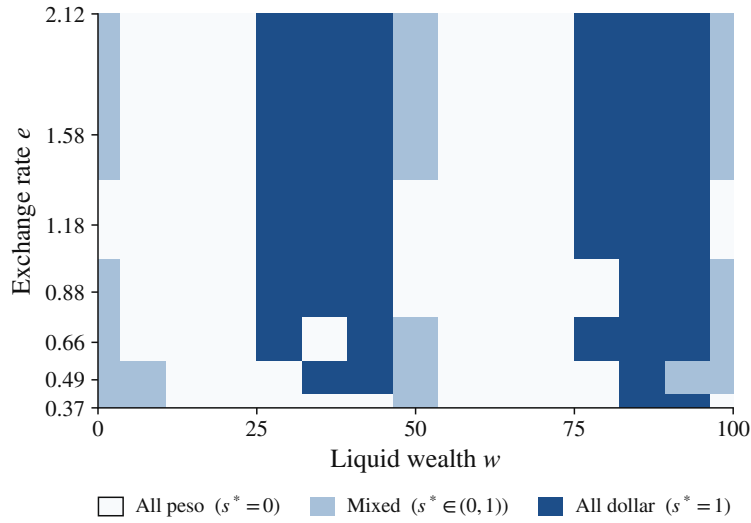


Figure 4: Portfolio dollarization decision in (w, e) space

Notes: The figure plots the optimal portfolio dollarization choice $s^*(w, e)$ as a function of liquid wealth w (horizontal axis) and the exchange rate e (vertical axis). White cells indicate the all-peso corner ($s^* = 0$); dark cells indicate the all-dollar corner ($s^* = 1$); gray cells indicate an interior solution ($s^* \in (0, 1)$). The predominantly binary pattern replicates the mass points at zero and one documented in the EFHU survey (Figure 1).

Figure 5 shows the exchange-rate response of the two main policy margins: the durable stock (panel a) and liquid savings (panel b), each comparing adjusting (solid) to non-adjusting households (dashed). In panel (a), the reset target d^* rises with the peso depreciation—a weaker exchange rate raises the peso value of dollar savings and income, relaxing the budget constraint at the adjustment margin—while the depreciated stock $(1 - \delta)d$ of non-adjusters is independent of e . In panel (b), both lines for savings rise with e , reflecting the balance-sheet appreciation of dollar assets, but adjusting households consistently carry higher savings than non-adjusters.

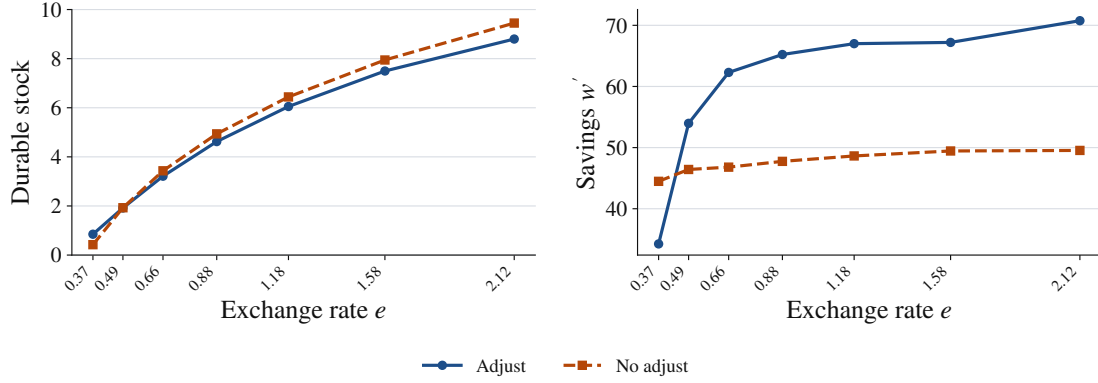


Figure 5: Durable stock and savings as a function of the exchange rate

Notes: Each panel plots the conditional mean policy as a function of the exchange rate e , averaging over all other state variables. Solid line: adjusting households. Dashed line: non-adjusting households. Panel (a): the solid line shows the optimal reset target d^* ; the dashed line shows the inherited stock $(1 - \delta)d$. Panel (b): next-period liquid savings w' .

Figure 6 characterizes the savings policy, plotting mean next-period liquid wealth w' against each state dimension. Adjusting households consistently carry *higher* end-of-period savings than non-adjusters—a selection effect: households adjust precisely when they have accumulated sufficient wealth to finance the purchase and still maintain liquid buffers. Panels (b) and (c) display standard precautionary patterns: savings rise steeply with current wealth, while the gap between lines widens with the inherited durable stock.

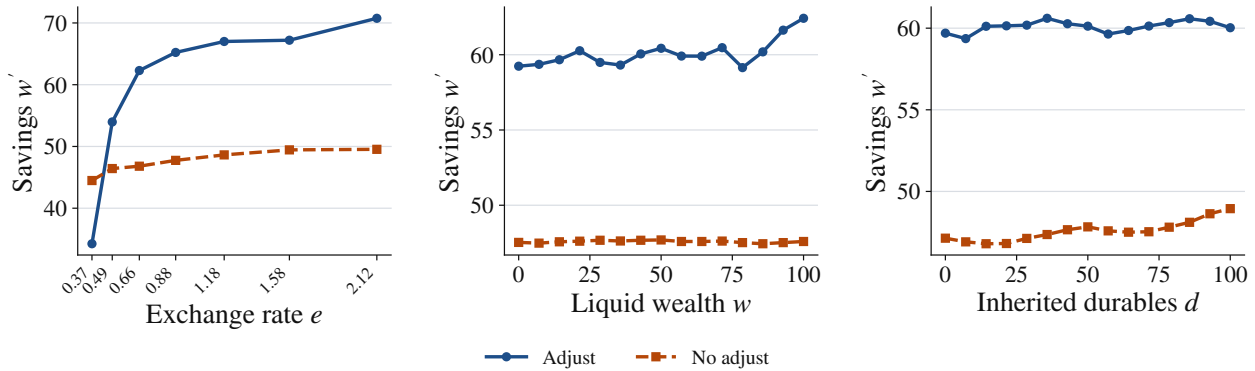


Figure 6: Savings policy by state dimension

Notes: Each panel plots the conditional mean of next-period liquid wealth w' as a function of one state variable, averaging over all other states. Solid line: households that adjust durables. Dashed line: households that do not adjust. Panel (a): savings by exchange rate e . Panel (b): savings by current liquid wealth w . Panel (c): savings by inherited durable stock d .

4 Quantitative Analysis

This section maps the model to Uruguayan data and describes how each parameter enters the targeted moments.

4.1 Calibration

I calibrate outside the model all parameters that are standard in the quantitative household literature or directly observable from aggregate data. I estimate by simulation the remaining parameters that govern portfolio choice and durable adjustment frictions in quarterly frequency.

I set $\beta = 0.986$, the inverse of the average quarterly real peso deposit return from BCU data, and $\gamma = 2$, a standard value in the quantitative household literature. These parameters facilitate comparability with the existing literature.

Idiosyncratic labor income follows an AR(1) process. I calibrate this block using harmonized Uruguay microdata from the Luxembourg Income Study (LIS) Database (Luxembourg Income Study (LIS), 2025). Specifically, I use person-level LIS files for 2004–2024 and estimate pseudo-panel wage dynamics on hourly wages in the main job for employees ages 25–55. The preferred specification uses only three-year cohort transitions and yields a quarterly persistence $\rho_y = 0.93$ and a reduced-form innovation standard deviation $\sigma_y = 0.20$. Relative to the Berger–Vavra benchmark ($\rho_y = 0.974$, $\sigma_y = 0.104$), the Uruguayan wage process is less persistent and more volatile, but it is substantially closer to the benchmark than calibrations based on household labor income.²

The nominal peso–dollar exchange rate follows an AR(1) process with persistence ρ_e and volatility σ_e . To calibrate the parameters I use the daily nominal UYU/USD spot rate from the Central Bank, converted to quarterly averages. Since the log nominal rate exhibits a persistent depreciation trend and a near-unit root in levels, I remove the trend component using the Hamilton 2018 filter and estimate an AR(1) on the residuals via OLS. The procedure yields $\rho_e \approx 0.85$ and $\sigma_e \approx 0.063$ (quarterly), which I round to the values in Table 3³.

I calibrate the non-durable utility share v using the 2016 *Encuesta de Gastos e Ingresos de los Hog-*

²The LIS exercise uses repeated cross-sections rather than a true panel, so the permanent-transitory decomposition is weakly identified. I therefore use the wage-based reduced-form estimate as the baseline calibration target and treat the exact split between persistent and transitory risk as a robustness issue.

³The Hamilton filter is preferred over the HP filter because it avoids endpoint bias and spurious low-frequency dynamics (Hamilton, 2018). As a robustness check, HP filtering with $\lambda = 1600$ gives similar values ($\rho_e \approx 0.72$, $\sigma_e \approx 0.054$).

ares (ENGIH 2016), Uruguay’s official household expenditure survey⁴. I set $\nu = 0.88$, consistent with the benchmark of Berger and Vavra (2015), using U.S. NIPA data, and Kaplan and Violante (2014), who use $\nu = 0.85$. Appendix B reports additional moments from ENGIH — including the durable investment rate and household savings rate — as non-targeted checks.

In the model, durable goods carry U.S. dollar prices with full pass-through to local-currency prices, consistent with the evidence documented in Section 2. The non-durable good is the numeraire. I set the risk-free return on local liquid assets to ensure consistency with the discount factor. The local interest rate r is the average real return on peso deposits, computed as the nominal deposit rate minus CPI inflation. The dollar interest rate $r^{\$}$ is the average real return on dollar deposits, computed as the nominal deposit rate minus U.S. CPI inflation. Both rates are at quarterly frequency and expressed in real terms.

Finally, I set the durable price parameter p_d to match the median durable value in EFHU relative to median quarterly income, which gives $p_d = 5.0$ in normalized units (Drenik and Perez, 2021). The parameter captures the typical cost of replacing durables relative to income and is central for quantifying the exchange-rate pass-through mechanism. In the SOE setting, durables are imported at world prices so p_d is exogenous; Appendix G discusses the closed-economy alternative where p_d is endogenous.

Table 3 summarizes the calibrated parameters and the estimated parameters.

⁴I aggregate monthly-equivalent household expenditure by division: furnishings and household equipment and transport including vehicles together account for 11.6% of consumption expenditure on average (median: 8.7%), implying a non-durable share of approximately 88%.

Table 3: Model parameters: calibration and estimation

Parameter	Description	Value	Source / target
<i>Calibrated</i>			
β	Discount factor	0.986	BCU data
γ	Risk aversion	2.00	Standard
δ	Durable depreciation	0.0125	Standard
r	Local interest rate	0.0142	BCU data
$r^{\$}$	Dollar interest rate	0.011	BCU data
ρ_y	Income persistence	0.93	LIS Uruguay
σ_y	Income innovation std. deviation	0.20	LIS Uruguay
ρ_e	Exchange-rate persistence	0.85	BCU data, Hamilton filter
σ_e	Exchange-rate std. deviation	0.063	BCU data, Hamilton filter
p_d	Durable price (USD)	5.0	Drenik and Perez (2021)
v	Non-durable utility share	0.88	ENGIH 2016
<i>Estimated by SMM</i>			
f	Resale haircut (irrevers.)	0.0225 (0.0006)	Duration + durable wealth mean
κ	Dollar portfolio wedge	0.0242 (0.5385)	Dollar share moment

Notes: Estimated parameters are reported with delta-method standard errors in parentheses. All parameters are at quarterly frequency. Income process (ρ_y, σ_y) estimated from LIS Uruguay microdata (own estimates; see calibration section). The Berger–Vavra benchmark ($\rho_y = 0.974$, $\sigma_y = 0.104$) uses U.S. data and is reported for comparison only. Exchange-rate parameters (ρ_e, σ_e) are estimated by OLS on the cyclical component of the quarterly log nominal UYU/USD rate, extracted using the Hamilton (2018) filter ($h = 8$ quarters). Interest rates are quarterly and expressed in real terms. The durable price is denominated in U.S. dollars.

4.2 Solution and simulation

The dynamic state is $S_t = (w_t, d_{t-1}, e_t, y_t)$, where w_t denotes liquid wealth, d_{t-1} the inherited stock of durables, and (e_t, y_t) follow exogenous AR(1) processes. I normalize all real variables by the median quarterly household income, improving numerical stability and facilitating interpretation.

I discretize liquid wealth on $[0, \bar{w}]$ and durables on $[0, \bar{d}]$ using an analogous grid. I set upper bounds $\bar{w}$ and $\bar{d}$ high enough to keep the probability of hitting the upper bound negligible. I discretize the exchange-rate and income processes using Tauchen’s method, matching the AR(1) specifications in (3.2)–(3.3). I construct the joint transition matrix for (e_t, y_t) as the Kronecker product of the individual Markov chains.

I solve the household problem with value function iteration. At each grid point, I evaluate both the adjustment and no-adjustment branches and take the maximum. Within each branch, the household chooses the optimal dollar share $s_t \in [0, 1]$ to maximize expected continuation value, accounting for

the state-contingent realization of next-period wealth under exchange-rate risk. Iteration continues until the maximum absolute change in the value function falls below 10^{-4} .

Given the converged policy functions, I simulate a panel of $N = 500$ households for $T = 1,500$ quarters, discarding the first 250 as burn-in. For each household-period, I record consumption, savings, portfolio composition, durable holdings, and the adjustment indicator. I compute simulated moments from the ergodic distribution of the simulated panel and compare them to their empirical counterparts.

4.3 Estimation

I estimate the remaining parameters using the Simulated Method of Moments (SMM). For a given parameter vector, I solve the household problem, simulate a large cross-section of households from the stationary distribution, and compute simulated moments. I choose parameters to minimize

$$\hat{\theta} = \arg \min_{\theta} \left(m^{\text{data}} - m^{\text{model}}(\theta) \right)' W \left(m^{\text{data}} - m^{\text{model}}(\theta) \right),$$

where W is a diagonal weighting matrix.

To minimize the SMM objective, I use a multistart random search. For each start, I draw parameter vectors uniformly from the admissible bounds, evaluate the objective function repeatedly, and retain the best candidate. I then repeat the search across many starts with different seeds and initial points, yielding tens of thousands of objective-function evaluations overall. Finally, I compute standard errors with the delta method, using a numerical Jacobian of simulated moments evaluated at the final estimate.

Since EFHU consists of repeated cross-sections rather than household panels, I compute all empirical moments at the cross-sectional level. Accordingly, I evaluate the model in its stationary distribution, and I construct simulated moments to mirror the survey moments.

The baseline estimation targets three moments that discipline the model's two central margins: durable adjustment and portfolio dollarization. Specifically, the targeted moments include:

1. the mean duration since last durable adjustment;
2. the mean durable share of total household wealth;
3. the mean dollar share of liquid assets.

Together, these moments identify the two estimated parameters. The resale haircut f is primarily disciplined by durable tenure and the mean durable wealth share, while the portfolio wedge κ is primarily disciplined by the dollar-share moment. With two parameters and three moments, the model is over-identified by one restriction.

4.4 Model fit

Table 4: Targeted moments: data and model

Moment	Data	Data SE	Model	Gap (Model–Data)
Mean years since last adjustment	23.178	(0.214)	20.069	−3.110
Mean durable share of wealth	0.829	(0.010)	0.462	−0.367
Mean dollar share of liquid assets	0.080	(0.004)	0.143	+0.063

Notes: Data moments are computed from pooled EFHU waves (2012, 2014, 2017) using survey weights. Standard errors in parentheses are computed by household-level block bootstrap (500 replications). “Years since last adjustment” is self-reported tenure since last housing or vehicle upgrade. “Dollar share” equals dollar deposits divided by total liquid assets (peso plus dollar deposits). “Durable share of wealth” equals the value of durables (housing plus vehicles at self-reported values) divided by total household wealth (durables plus liquid assets). Model moments are computed from the stationary distribution of 500 simulated households over 1,500 quarters (250-quarter burn-in), evaluated at the current estimate $(\hat{f}, \hat{\kappa}) = (0.0225, 0.0242)$.

Table 4 compares the targeted empirical moments to their simulated counterparts. The fit is mixed. The model somewhat under-predicts duration (20.1 vs. 23.2 years), materially under-predicts the durable wealth share (0.462 vs. 0.829), and overshoots the dollar share (0.143 vs. 0.080 in the data), though the absolute gap on the dollar share is smaller than on the durable share.

Parameter-level inference indicates asymmetric identification strength across frictions. The estimate of f is tightly estimated at $\hat{f} = 0.0225$ with std. error = 0.0006, so the durable-adjustment friction is precisely pinned down in the baseline three-moment specification. By contrast, κ is weakly identified: $\hat{\kappa} = 0.0242$ with std. error = 0.5385, implying that the data provide very little curvature on the dollar-saving wedge under the baseline target set. The estimate of f also lies below the Berger-Vavra range (0.043–0.068), but the comparison is not one-for-one. In this model, households jointly choose durable adjustment and currency composition, so the effective friction is not captured by f alone. Part of the adjustment friction that would appear as a higher fixed cost in a one-asset durable model can instead be absorbed by the interaction of a lower f with a positive portfolio wedge κ . The pattern is consistent with the moment mapping: durable-adjustment moments strongly discipline f , while a single dollar-share moment provides limited curvature for

κ . The over-identifying restriction is not rejected at conventional levels under the current specification ($J \approx 0.286$, $p \approx 0.593$, with one over-identifying degree of freedom). Since κ is weakly identified, policy experiments that move dollar-saving costs are more uncertain than experiments that operate through durable adjustment costs.

Two implications of the fit deserve emphasis before proceeding to the counterfactuals. First, the gap in the durable wealth share (0.46 vs. 0.83) suggests the model understates the importance of durables in household balance sheets. If households actually hold durables more intensively than the model implies, welfare costs associated with durable frictions and peso pricing are likely understated: the mechanism relies on the durable adjustment margin, so understating durable intensity biases the estimated CEV toward zero. Second, the asymmetry in identification sharpens the hierarchy of findings: the peso-pricing counterfactual ($\text{CEV} \approx -11\%$) is the most precisely identified result because it operates primarily through f , which is tightly estimated; the dollar-access counterfactual ($\text{CEV} \approx -9\%$) should be read as indicative given weak identification of κ .

The model also matches the qualitative empirical correspondence that motivates the framework. It reproduces limited participation with high conditional dollarization (Fact 1), a positive income gradient in dollar holdings (Fact 3), and infrequent durable adjustment (Fact 4). Fact 2 (dollar pricing of durables) is imposed as an institutional feature rather than estimated from moments, so the quantitative fit should be interpreted as conditional on that pricing environment. The main mismatch remains the level of durable intensity in household balance sheets, discussed above.

4.5 Generalized Impulse Response Functions

Since the model features discrete adjustment decisions and nonlinear policy functions, I compute generalized impulse response functions (GIRFs) following Koop et al. (1996). The GIRF to an exchange-rate shock is defined as

$$\text{GIRF}(h) = \mathbb{E}[x_{t+h} \mid \varepsilon_t^e = \Delta] - \mathbb{E}[x_{t+h}],$$

where expectations are taken over the ergodic distribution of initial states and the realizations of future shocks. I implement the simulation by simulating 500 histories, each starting from a state drawn from the stationary distribution. For each history, I compare the path following a depreciation shock to the path without the shock, holding the sequence of future income and exchange-rate innovations fixed. The reported GIRF is the average difference across histories, with confidence bands reflecting simulation uncertainty.

Figure 7 reports probability-scaled GIRFs for a one-standard-deviation exchange-rate depreciation under the estimated process ($\rho_e = 0.85$, $\sigma_e = 0.063$ quarterly). A one-standard-deviation innovation is therefore a moderate shock, about 6.3 log points (roughly a 6.5% depreciation), not a full crisis scenario. The plotted paths equal the conditional GIRF multiplied by a quarterly arrival probability of 0.10, so the figure should be read as an expected-flow response rather than a pure conditional shock path. Each line is the difference between the shocked path and an otherwise identical no-shock path with the same future innovations, averaged over initial states from the ergodic distribution. Relative to that no-shock benchmark, consumption falls on impact, durables fall, liquid wealth jumps, and the dollar share declines.

The impact responses reflect the model's balance-sheet mechanics. The depreciation raises the peso replacement cost of durables, which leads households to defer adjustment and redirect resources toward precautionary saving rather than current spending, so non-durable consumption falls on impact. At the same time, the peso value of existing dollar assets rises, which boosts measured liquid wealth. The bottom-left panel plots the exogenous shock itself: it decays geometrically at rate ρ_e , so it is persistent but stationary. The endogenous responses can look long-lived because the shock hits stock variables (wealth, durable stock, portfolio composition) and adjustment is lumpy; these stocks re-optimize gradually, so differences can remain visible over a 40-quarter window even in a stationary model. In magnitude, the responses are small-to-moderate in expected-flow units and large in pure conditional units. In the current baseline runs, the probability-scaled ($\pi_q = 0.10$) impact response is about -0.9% for consumption in log terms.

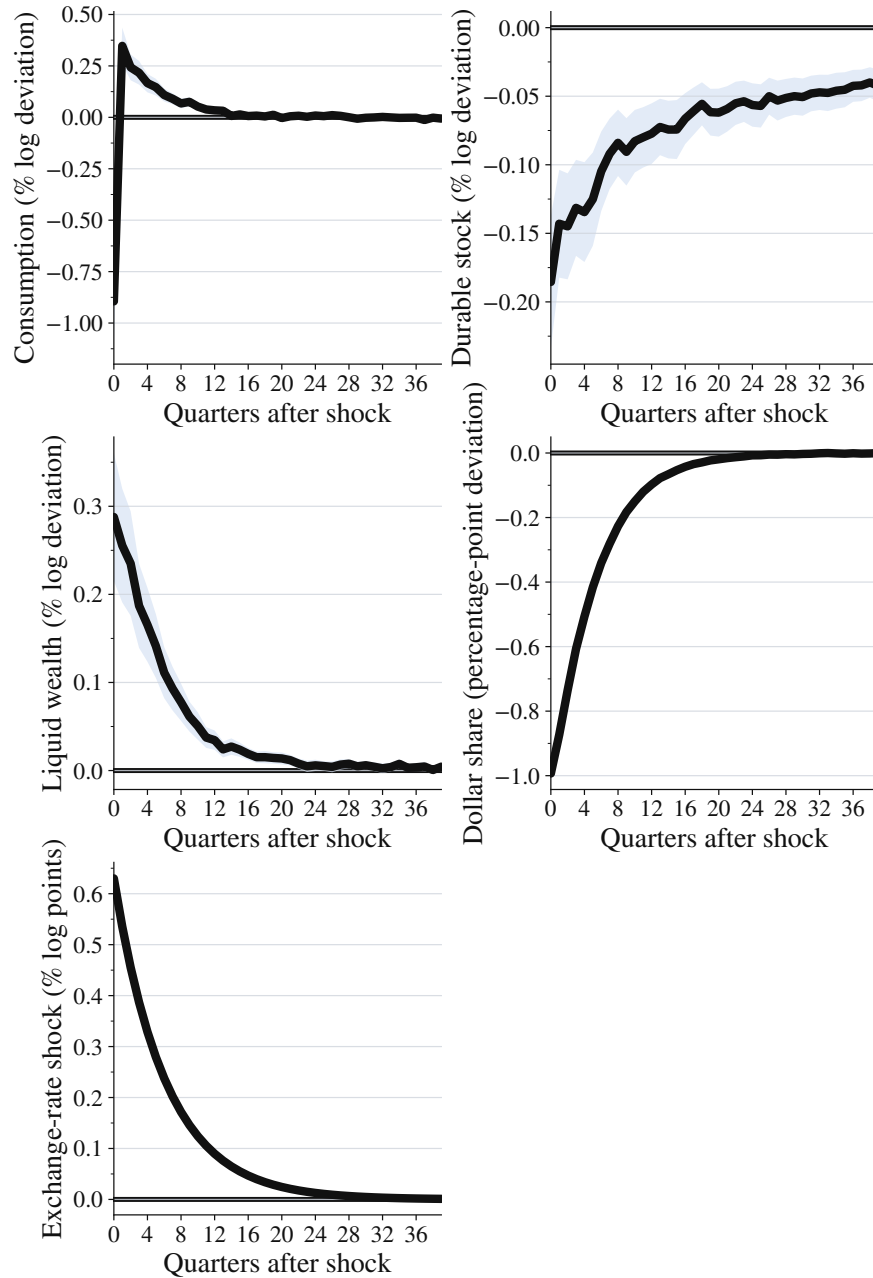


Figure 7: Impulse Response Functions

Notes: Probability-scaled generalized impulse response functions to a one-standard-deviation exchange-rate depreciation, computed following Koop et al. (1996). The plotted response equals the conditional GIRF multiplied by a quarterly arrival probability of 0.10. The shock panel reports the exogenous exchange-rate shock path in percentage log points; the consumption, durables, and liquid-wealth panels are reported in percentage log deviations; the dollar-share panel is reported in percentage-point deviations. Each response is the average difference between shocked and no-shock paths using common random numbers for future innovations. The sample includes 5,000 households per history and 500 histories drawn from the ergodic distribution (more households per history than the estimation panel to reduce simulation noise in GIRF averages). Shaded bands are 95% simulation bands computed across 500 histories drawn from the ergodic distribution; they reflect simulation uncertainty, not empirical sampling variation. Horizontal axis: quarters after shock.

The GIRF dynamics, with deferred durable adjustment, temporary liquid-buffer accumulation, and short-lived dollar-share responses, motivate the welfare experiments in Section 5. They isolate the margins through which exchange-rate shocks propagate in the estimated model and therefore identify the frictions that matter most for welfare. Appendix I complements this baseline analysis with a crisis-process GIRF benchmark that maps the model to Uruguay’s 2002 financial crisis, using the devaluation magnitude documented by Drenik et al. (2018).

5 Counterfactuals

I use counterfactual experiments to isolate the mechanisms through which (i) nonconvex durable adjustment and (ii) access to dollar saving shape portfolio choice, expenditure dynamics, and welfare. I focus the main quantitative comparison on a core set of stationary counterfactuals. In each exercise, I change one parameter while holding all other parameters fixed at their estimated values. The objective is not just household-level welfare accounting: the experiments clarify how deposit and pricing dollarization shape the transmission of currency shocks to household demand.

Let $V^{\text{base}}(S)$ and $V^{\text{cf}}(S)$ denote the value functions in the baseline and counterfactual economies. Let μ^{base} denote the stationary distribution of states in the baseline. I summarize welfare by the stationary-average value,

$$\bar{V}^j \equiv \int V^j(S) d\mu^{\text{base}}(S), \quad j \in \{\text{base}, \text{cf}\},$$

and compute the consumption-equivalent variation (CEV),

$$\lambda = \left(\frac{\bar{V}^{\text{cf}}}{\bar{V}^{\text{base}}} \right)^{\frac{1}{1-\gamma}} - 1. \quad (5.1)$$

With $\gamma > 1$ and value functions negative, $\lambda > 0$ denotes a welfare *gain* in the counterfactual relative to the baseline; $\lambda < 0$ denotes a welfare loss (Lucas, 1987). Throughout, welfare comparisons hold fixed the baseline stationary distribution, so changes reflect pure policy-function effects rather than compositional shifts.

To benchmark the effects, I compare them to standard welfare estimates. The welfare cost of U.S. business cycles is commonly estimated at 0.05–0.1% of consumption (Lucas, 1987; Imrohoroğlu, 1989), while gains from trade liberalization for small open economies are often in the range of 1–5% (Arkolakis et al., 2012; Caliendo and Parro, 2015). Table 5 reports the core stationary counterfactuals and volatility margins. The first column reports the CEV relative to the baseline;

the second and third columns report the percentage change in the volatility of durable expenditure and non-durable consumption, respectively, relative to the baseline. The table shows that durable adjustment frictions have a larger welfare cost than dollar pricing or access, and that the welfare cost of dollar pricing is primarily driven by the loss of insurance rather than price volatility.

5.1 Durable adjustment frictions

I increase the resale haircut from its estimated value of $f = 0.0225$ to $f = 0.043$, the lower end of the Berger-Vavra range, to discipline the magnitude of the friction. The baseline parameter matches observed replacement frequencies in the data; the counterfactual raises irreversibility to levels comparable to high transaction-cost environments or thin secondary markets for durable goods (Lacetera et al., 2012; Berger and Vavra, 2015; Piazzesi and Schneider, 2016).

Relative to the baseline, higher adjustment costs reduce the frequency of durable replacement and lengthen spells between updates. Conditional on adjustment, households undertake larger discrete changes in the durable stock. The increased rigidity raises the value of liquid buffers and tilts portfolios toward assets that hedge the dollar-denominated replacement cost of durables.

Quantitatively, the counterfactual implies a CEV of -20.99% , the largest welfare loss among the core stationary experiments. Relative to baseline, durable-expenditure volatility drops sharply from about 3.83 to 0.402, and consumption volatility declines from 0.549 to 0.380 (Table 5). The reduction in $\sigma(\Delta d)$ reflects a lower frequency of large durable updates⁵.

5.2 Dollar pricing as a natural hedge: local-currency pricing counterfactual

The counterfactual directly addresses the paper’s central question: what is the welfare consequence of durable goods being denominated in dollars? I replace dollar pricing with local-currency pricing, so the peso price of durables stays fixed and does not respond to exchange-rate movements. Dollar savings continue to carry exchange-rate risk. I hold all other parameters at their estimated values.

A CEV of -10.66% suggests households would be worse off when durables are peso-priced. The negative sign is counterintuitive at first glance. One might expect insulating durable prices from exchange-rate fluctuations to help households. The mechanism runs the other way.

⁵Setting δ arbitrarily close to one nearly shuts down the durable channel and produces a much larger loss. Once durable adjustment effectively disappears, the welfare consequences become dramatically larger, which underscores that durables are central to the model’s transmission.

Under dollar pricing, the local-currency cost of durable replacement and the local-currency value of dollar savings both increase proportionally when the peso depreciates. The durable price satisfies $P_t^d = e_t p_d$, while the peso return on dollar savings includes the term $(1 + r^{\$}) \frac{e_{t+1}}{e_t}$. Both quantities scale with e_t , so they move together. As a result, the household's effective real cost of durable replacement—measured in units of dollar savings—is partially insulated from exchange-rate movements. A depreciating peso raises durable prices, but it simultaneously raises the peso value of dollar assets by the same factor.

With peso pricing, the covariance disappears. Durable prices no longer rise with a depreciation, but neither does the real cost of durable adjustment fall with an appreciation. More importantly, the motive to hold dollar assets is weakened: dollars no longer serve as a durable-price hedge, because the durable price is not dollar-denominated. As a result, households shift toward peso assets, losing the hedging channel. The loss of the natural hedge is larger than the gain from smoothing durable prices, producing a net welfare loss.

The volatility moments in Table 5 also align with the mechanism. Relative to baseline, durable-expenditure volatility remains lower (3.83 to 1.691), but consumption volatility rises (0.549 to 0.571). This pattern is consistent with weaker currency hedging in liquid portfolios once durables are no longer dollar-priced.

Two implications follow for policy. First, simply pegging or fixing durable prices in local currency does not necessarily improve household welfare if it destroys the hedging complementarity between dollar assets and dollar-priced durables. Second, the primary welfare cost of dollar denomination is not the price volatility it creates, but the insurance it enables—a distinction that is invisible to frameworks that abstract from household portfolio choice. More broadly, the experiment shows that pricing conventions can reshape the household side of exchange-rate transmission by changing whether a depreciation primarily tightens real replacement costs or also activates a balance-sheet hedge.

5.3 Restrictions on dollar saving

I increase the portfolio wedge from its estimated baseline value of $\kappa = 0.0242$ to $\kappa = 0.05$, roughly a two-fold increase in the proportional cost of foreign-currency saving. Relative to the baseline, a higher κ reduces equilibrium dollarization and weakens households' ability to hedge the dollar price of durable replacement.

Portfolios shift toward peso assets, increasing exposure to exchange-rate movements. The experi-

ment approximates regimes of capital controls, or severe financial repression, such as Argentina’s 2001–2002 (Reinhart and Sbrancia, 2015). The magnitude effectively shuts down the hedging channel while preserving model structure. In policy terms, it captures the household-side consequence of reducing deposit dollarization: a weaker private hedge against currency movements.

The implied welfare loss of 9.58% would be economically large and comparable to standard gains from trade, underscoring the importance of the dollar hedging channel: when households cannot dollarize their portfolio, they would lose the principal instrument for insuring the dollar price of durable replacement. Relative to baseline, both volatility margins increase, with $\sigma(\Delta d)$ rising from 3.83 to 4.446 and $\sigma(\log c)$ rising from 0.549 to 0.612. Limiting access to dollar saving therefore generates first-order welfare costs. Thus, interventions that reduce household foreign-currency saving may lower system-wide dollarization, but they do so by shifting exchange-rate risk back onto household spending and durable-adjustment margins.

Table 5: Core stationary counterfactuals: volatility diagnostics

Counterfactual	CEV (%)	$\sigma(\Delta d)$	$\sigma(\log c)$
Baseline	—	3.827	0.549
Higher durable adj. cost	−20.99	0.402	0.380
Peso-priced durables	−10.66	1.691	0.571
Restricted dollar saving ($\kappa = 0.05$)	−9.58	4.446	0.612

Notes: $\sigma(\Delta d)$ is the standard deviation of net durable investment $d_t - d_{t-1}(1 - \delta)$ in the counterfactual stationary distribution. $\sigma(\log c)$ is the standard deviation of log non-durable consumption in the counterfactual stationary distribution.

6 Robustness, Extensions, and Discussion

6.1 Interaction between Dollar Access and Adjustment Frictions

The preceding experiments evaluate welfare losses from changing one friction at a time, starting from an economy in which households have full access to dollar saving at the estimated cost κ . A complementary question is whether the costs are amplified or attenuated when dollar access is already restricted. If households cannot dollarize, they lose a key hedging instrument. Currency risk and durable illiquidity may then become considerably more damaging as the insurance channel is blocked.

To examine the issue, I construct a restricted baseline economy in which I hold all parameters at their estimated values except the portfolio wedge, which I raise to $\kappa = 0.05$ (from the estimated $\kappa = 0.0242$). At the chosen level, the cost of holding dollar assets is substantially elevated, and equilibrium dollarization falls sharply. The calibration approximates an economy with moderate capital controls or financial repression—policies observed in several emerging markets (Reinhart and Sbrancia, 2015). I hold all other parameters, including the exchange-rate process, durable prices, and income dynamics, constant so the restricted baseline and the estimated baseline differ only in household access to dollar saving.

Starting from the restricted baseline, I apply the same core counterfactuals as in Section 5: higher durable adjustment costs and peso-priced durables. Table 6 reports the results and compares them to the corresponding baseline experiments.

Table 6: Welfare and volatility effects under restricted dollar access ($\kappa = 0.05$)

Counterfactual	Baseline ($\kappa = 0.0242$)			Restricted baseline ($\kappa = 0.05$)		
	CEV (%)	$\sigma(\Delta d)$	$\sigma(\log c)$	CEV (%)	$\sigma(\Delta d)$	$\sigma(\log c)$
Higher durable adj. cost	−20.99	0.402	0.380	−21.86	0.414	0.465
Peso-priced durables	−10.66	1.691	0.571	−14.88	1.424	0.714

Notes: Each row reports the CEV (%) and volatility measures in the counterfactual economy, relative to the respective baseline. Left panel (baseline, $\kappa = 0.0242$): baseline counterfactual benchmarks. Right panel (restricted baseline, $\kappa = 0.05$): the starting economy has substantially elevated dollar saving costs; all other parameters held at estimated values. $\sigma(\Delta d)$ is the standard deviation of net durable investment $d_t - d_{t-1}(1 - \delta)$; $\sigma(\log c)$ is the standard deviation of log non-durable consumption.

The comparison across columns is informative about the interaction between dollar access and durable illiquidity. Since f and κ are orthogonal in the model, differences in the welfare cost of higher f across columns reflect indirect effects operating through portfolio re-optimization rather than a direct channel; Appendix H.2 reports additional welfare-measure decompositions for the counterfactuals.

In the updated runs, the higher adjustment cost row shows mild amplification: the welfare loss moves from -20.97% in the estimated baseline to -21.86% under restricted dollar access. When dollar saving is already expensive, households have weaker hedging capacity, so additional durable illiquidity is more costly rather than less. In this calibration, the two frictions act as complements.

The peso-pricing row also shows amplification: switching durables from dollar to peso denomination is more costly when dollar access is already restricted (CEV falls from -10.62% to -14.88%), consistent with a weaker hedging margin in the restricted baseline.

6.2 General-equilibrium robustness

The partial-equilibrium baseline is natural for a small open economy: Uruguay's exchange rate and world interest rate r^S are externally determined, so the prices households face are exogenous by assumption. Nonetheless, a general equilibrium (GE) exercise is useful to ask whether endogenizing the domestic peso rate r and the wage w materially changes welfare magnitudes.

I implement a GE exercise following Aiyagari (1994): bisection over r to find the stationary equilibrium in which aggregate peso deposits equal firm capital demand, with household-side parameters held fixed. The domestic asset-market clearing condition is

$$\underbrace{\int (1 - s^*(S)) w'^*(S) d\mu^*(S)}_{\text{aggregate peso deposits}} = \underbrace{K^*}_{\text{firm capital demand}},$$

where $S = (w, d, e, y)$ is the household state, $s^*(S) \in [0, 1]$ is the optimal dollar portfolio share, $w'^*(S)$ is optimal next-period liquid wealth, μ^* is the stationary distribution, and K^* satisfies the firm's first-order condition $r^* + \delta_K = \alpha Z(K^*)^{\alpha-1}$. Dollar deposits ($s^* w'^*$) are net foreign assets that clear in the world capital market; durable purchases are dollar-priced imports that also clear abroad. No separate domestic market clearing is required for either.

A key distinction from Berger and Vavra (2015) is the treatment of durable investment. In their closed-economy framework the resource constraint is $Y = C + I_k + I_d$, so lumpy durable adjustment in I_d can be absorbed by the liquid capital margin I_k without amplifying C . In this SOE, durable goods are dollar-priced imports: households finance durable purchases out of dollar savings, and durable investment clears in the world goods market. The domestic resource constraint therefore reduces to $Y = C + I_k$. The same buffering logic applies—exchange-rate-induced durable reallocation is absorbed through the peso savings margin and the current account rather than through domestic capital—but the GE closing condition involves only the domestic peso asset market. In this sense, the GE absorption mechanism here is closer in spirit to Khan and Thomas (2008) and Bachmann et al. (2013), who show that lumpy firm-level investment is largely neutralized by interest-rate adjustment in GE, than to Berger and Vavra's richer closed-economy setting where an additional quantity channel—the reallocation between I_k and I_d —is available to absorb lumpy durable spending.

The bisection converges to $r^* = 4.20\%$ (versus $r_{PE} = 1.42\%$) and $w^* = 0.740$ (versus $w_{PE} = 1.000$), with aggregate peso deposits $A^* = 5.28$ and domestic capital demand $K^* = 6.22$. Two opposing channels drive the attenuation of PE welfare costs. First, the higher equilibrium peso return raises the reward to domestic saving, which partially substitutes for dollar hedging and reduces the

marginal welfare cost of losing dollar pricing or dollar access. Second, the lower wage reduces labor income and shrinks the consumption scale of all households, compressing the stakes of every welfare comparison relative to PE. Together, these price adjustments absorb most of the partial-equilibrium incidence. The GE CEVs for the three core counterfactuals are near zero: -0.54% for higher adjustment costs (-21.0% in PE), -0.22% for peso-priced durables (-10.7% in PE), and -0.06% for restricted dollar saving (-9.6% in PE). PE magnitudes are therefore best read as an upper bound. The residual GE effects range from 0.06% (dollar-access restriction, essentially at the Lucas 1987 business-cycle benchmark of $\approx 0.05\%$) to 0.54% (adjustment frictions, comparable to welfare costs of moderate business-cycle risk under incomplete markets (Imrohoroğlu, 1989)). These are economically non-trivial: denomination frictions impose persistent welfare costs even after full domestic price adjustment. Table 7 summarizes the PE–GE comparison.

Table 7: Partial- vs. general-equilibrium comparison

	Partial equilibrium	General equilibrium
<i>Factor prices</i>		
Peso interest rate r	1.42%	4.20%
Wage w	1.000	0.740
<i>Welfare (CEV, %)</i>		
Higher adj. cost ($f = 0.043$)	-21.0	-0.54
Peso-priced durables	-10.7	-0.22
Restricted dollar saving ($\kappa = 0.05$)	-9.6	-0.06

Notes: Household parameters ($\hat{f}, \hat{\kappa}$) held fixed at PE estimates in both columns. GE closes only the domestic peso asset market; dollar savings and durable imports clear in world markets. See Appendix G for full diagnostics.

Two caveats apply. First, the strong dampening is partly an artifact of the SOE assumption: dollar prices, dollar savings, and the exchange rate all remain exogenous, so only the domestic peso rate and the wage adjust; in a richer open-economy setting, GE costs would be larger. Second, household parameters are held fixed at their PE estimates; re-estimation at GE prices could shift both parameter values and welfare conclusions. Appendix G reports the full decomposition and dynamic diagnostics.

6.3 Measurement and re-estimation robustness

Robustness checks on data construction and estimation are summarized briefly here and documented in detail in the appendix. On the measurement side, Appendix A shows that the baseline

moment construction uses survey weights, harmonized definitions across waves, and conservative sample restrictions aligned with model objects (e.g. portfolio shares computed among households with positive liquid assets). It also reports that wave-specific winsorization and the use of Wave 2 for the continuous currency-share variable are comparability choices, not favorable-sample choices. Appendix B provides external validation through ENGIH moments used to discipline v .

On the estimation side, alternative filters for the exchange-rate process and alternative simulation/seed configurations deliver similar qualitative conclusions (Appendix D). I additionally run a five-moment augmented re-estimation that adds dollar-share volatility and dollar participation to the baseline three moments. The augmented run leaves the qualitative mechanism unchanged and mainly sharpens discipline on the dollarization margin; details are reported in Appendix D.1.

These checks support the baseline interpretation: the durable-friction margin is comparatively well identified, while the dollar-saving wedge remains the weaker dimension of the specification.

7 Conclusion

This paper shows that dollar pricing of durable goods is welfare-improving in partially dollarized small open economies. When durable replacement costs and dollar savings share a common denomination, the household balance sheet provides a natural hedge: both sides appreciate with the exchange rate, providing insurance when it is most needed. Removing dollar pricing by switching to local-currency denomination would sever this hedge and generate a model-implied welfare loss of about 11%. The result reframes the policy question: the concern is not dollar pricing per se, but ensuring households can access the dollar assets that make dollar pricing beneficial.

The counterfactuals underscore the hedging channel. Restricting dollar access implies a welfare cost near 9%, while higher durable adjustment frictions generate a welfare loss above 20%—both larger than standard trade liberalization estimates. These magnitudes arise from the same mechanism block highlighted in the model: adjustment frictions, portfolio hedging trade-offs, state-dependent precaution, and the interaction between portfolio composition and adjustment timing.

The general-equilibrium extension confirms that partial-equilibrium welfare magnitudes are best read as upper bounds. Endogenizing the domestic peso rate and wage via an Aiyagari-style market-clearing condition—while keeping dollar savings and durable investment in world markets—raises r from 1.4% to 4.2% and lowers w by 26%. These factor-price adjustments absorb most of the welfare incidence visible in PE; the residual GE effects—ranging from 0.06% to 0.54% across counterfactuals—span the Lucas (1987) representative-agent benchmark and the incomplete-markets

range of Imrohoroglu (1989), remaining positive and economically interpretable. Even under this more conservative benchmark, the welfare impacts of dollar pricing and dollar access are sizable and robust.

The model fits durable duration closely but under-predicts the durable wealth share and overshoots the dollar share, reflecting the tight parameterization of the portfolio block. Inference indicates κ is more weakly identified than f , so welfare estimates for dollar-access restrictions carry greater uncertainty than those for adjustment frictions.

The results suggest that household dollarization is an endogenous response to pricing frictions and exchange-rate risk in partially dollarized economies. Policies aimed at reducing dollarization face a fundamental trade-off: while they may enhance systemic financial stability, they shift exchange-rate risk onto households and remove the hedging instrument that makes dollar-priced durables affordable. An alternative approach is to reduce the underlying demand for dollar hedging by promoting credible monetary policy and developing deep local-currency financial markets, rather than restricting the supply of dollar assets.

References

- Aiyagari, S. R. (1994). Uninsured idiosyncratic risk and aggregate saving. *Quarterly Journal of Economics*, 109(3):659–684.
- Arkolakis, C., Costinot, A., and Rodríguez-Clare, A. (2012). New trade models, same old gains? *American Economic Review*, 102(1):94–130.
- Bachmann, R., Caballero, R. J., and Engel, E. M. R. A. (2013). Aggregate Implications of Lumpy Investment: New Evidence and a DSGE Model. *American Economic Journal: Macroeconomics*, 5(4):29–67.
- Baley, I. and Blanco, A. (2021). Aggregate Dynamics in Lumpy Economies. *Econometrica*, 89(3):1235–1264.
- Barro, R. J. (2006). Rare disasters and asset markets in the twentieth century*. *The Quarterly Journal of Economics*, 121(3):823–866.
- Beraja, M. and Wolf, C. K. (2021). Demand Composition and the Strength of Recoveries. NBER Working Papers 29304, National Bureau of Economic Research, Inc.
- Berger, D. and Vavra, J. (2015). Consumption dynamics during recessions. *Econometrica*, 83(1):101–154.

- Caballero, R. J. (1993). Durable Goods: An Explanation for Their Slow Adjustment. *Journal of Political Economy*, 101(2):351–384.
- Caballero, R. J. and Engel, E. M. R. A. (1993). Microeconomic Adjustment Hazards and Aggregate Dynamics*. *The Quarterly Journal of Economics*, 108(2):359–383.
- Caliendo, L. and Parro, F. (2015). Estimates of the trade and welfare effects of nafta. *The Review of Economic Studies*, 82(1 (290)):1–44.
- Calvo, G. and Vegh, C. (1992). Currency Substitution in Developing Countries: An Introduction. Mpra paper, University Library of Munich, Germany.
- Castellares, R. and Toma, H. (2020). Effects of a mandatory local currency pricing law on the exchange rate pass-through. *Journal of International Money and Finance*, 106:102186.
- Cravino, J. and Levchenko, A. A. (2017). The Distributional Consequences of Large Devaluations. *American Economic Review*, 107(11):3477–3509.
- De Rosa, M. (2019). Wealth accumulation and its distribution in uruguay: first estimates of the untold half of the story. Working Paper BCU 2019-12, Banco Central del Uruguay, Montevideo.
- Drenik, A., Pereira, G., and Perez, D. J. (2018). Wealth Redistribution after Exchange Rate Devaluations. *AEA Papers and Proceedings*, 108:552–556.
- Drenik, A. and Perez, D. J. (2021). Domestic price dollarization in emerging economies. *Journal of Monetary Economics*, 122:38–55.
- Drenik, A. and Perez, D. J. (2024). Dollar Pricing, Inflation, and Search Frictions. Revise and resubmit, *Journal of Monetary Economics*.
- Hamilton, J. D. (2018). Why you should never use the Hodrick-Prescott filter. *Review of Economics and Statistics*, 100(5):831–843.
- Heathcote, J., Storesletten, K., and Violante, G. L. (2008). Insurance and opportunities: A welfare analysis of labor market risk. *Journal of Monetary Economics*, 55(3):501–525.
- Imrohoroglu, A. (1989). Cost of business cycles with indivisibilities and liquidity constraints. *Journal of Political Economy*, 97(6):1364–1383.
- Kaplan, G., Mitman, K., and Violante, G. L. (2020). The housing boom and bust: Model meets evidence. *Journal of Political Economy*, 128(9):3285–3345.

- Kaplan, G. and Violante, G. L. (2014). A model of the consumption response to fiscal stimulus payments. *Econometrica*, 82(4):1199–1239.
- Kehoe, T. and Nicolini, J. P. (2021). *A Monetary and Fiscal History of Latin America, 1960–2017*. University of Minnesota Press.
- Khan, A. and Thomas, J. K. (2008). Idiosyncratic Shocks and the Role of Nonconvexities in Plant and Aggregate Investment Dynamics. *Econometrica*, 76(2):395–436.
- Koop, G., Pesaran, M., and Potter, S. M. (1996). Impulse response analysis in nonlinear multivariate models. *Journal of Econometrics*, 74(1):119–147.
- Lacetera, N., Pope, D. G., and Sydnor, J. R. (2012). Heuristic thinking and limited attention in the car market. *American Economic Review*, 102(5):2206–36.
- Levy Yeyati, E. and Ize, A. (2005). Financial De-Dollarization; Is it for Real? IMF Working Papers 05/187, International Monetary Fund.
- Lucas, R. E. (1987). *Models of business cycles*. Yrjö Jahnsson lectures. B. Blackwell.
- Luxembourg Income Study (LIS) (2025). Luxembourg income study (lis) database. <https://www.lisdatacenter.org>. Harmonized microdata accessed via LISSY; Luxembourg, asbl.
- McKay, A. and Wieland, J. F. (2021). Lumpy durable consumption demand and the limited ammunition of monetary policy. *Econometrica*, 89(6):2717–2749.
- Piazzesi, M. and Schneider, M. (2016). Housing and macroeconomics. *Handbook of macroeconomics*, 2:1547–1640.
- Reinhart, C. M. and Sbrancia, M. B. (2015). The liquidation of government debt. *Economic Policy*, 30(82):291–333.
- Salomao, J. and Varela, L. (2022). Exchange rate exposure and firm dynamics. *The Review of Economic Studies*, 89(1):481–514.
- World Bank (2024). World development indicators. World Bank Open Data, <https://databank.worldbank.org/source/world-development-indicators>. Accessed 2024.
- Zhou, H. (2022). Open economy, redistribution, and the aggregate impact of external shocks. University of Princeton, Job Market Paper.

Appendix

The appendix is organized as follows. Sections A–B cover data construction and supplementary expenditure moments. Sections C–D document computation and estimation, including an augmented five-moment re-estimation. Sections E–F provide model intuition and the convex benchmark. Section G presents the GE robustness and closed-economy comparison. Section H reports additional welfare robustness. Section I reports the crisis-process GIRF benchmark, and Section J reports comparative statics.

A Data Appendix: EFHU Construction and Measurement

This section describes the Uruguayan Household Financial Survey (EFHU), the sample and variable definitions used to build empirical moments, and the measurement conventions that align the data with the model.

Survey structure. EFHU is a nationally representative household financial survey administered in Uruguay in three waves between 2012 and 2017. The survey is a repeated cross-section rather than a panel. It reports detailed household balance-sheet information, including liquid assets by currency, durable holdings, debts, and self-reported values for major assets. The currency breakdown of liquid assets is the feature that makes EFHU especially useful for studying household dollarization.

Sample construction. I pool the available waves and use household expansion factors throughout. For moments involving portfolio shares, I restrict the sample to households with positive liquid financial assets so the dollar share is well defined. For moments involving durable holdings, I use housing and vehicle values as the empirical counterpart to the model’s composite durable stock. For duration moments, I use the self-reported time since the last housing or vehicle upgrade.

Currencies and timing. I express all monetary variables in U.S. dollars. For each EFHU wave w conducted in calendar year t , I convert peso-denominated variables X_{it}^{UY} using the contemporaneous nominal exchange rate e_t (pesos per dollar):

$$X_{it}^{USD} = \frac{X_{it}^{UY}}{e_t}.$$

The convention aligns the data with the model’s pricing structure, in which durable goods are priced in dollars and have local-currency prices $P_t^d = e_t p_d$. All empirical moments used for estimation are ratios or shares and are therefore invariant to the choice of numeraire.

Income concept. To match the model’s normalization of durable holdings by income, I use labor income only. I construct quarterly labor income as

$$y_{it}^{\text{lab}} \equiv 3 \times \text{monthly labor income},$$

winsorized by wave to limit the influence of extreme values. All moments involving durable-to-income ratios in the data use y_{it}^{lab} as the denominator, and I construct simulated moments using the same definition.

Durable definitions. In the baseline specification, I define the household durable stock as the sum of housing and vehicle values. The distinction is relevant because depreciation rates and resale frictions differ across components, while the model abstracts from component-level heterogeneity by using a single composite durable good.

Wave alignment. Not all variables are measured equally across waves. In particular, the continuous currency-share variable that maps most closely into the model is available in Wave 2, so the baseline dollar-share target uses EFHU 2013 for direct comparability. Other balance-sheet moments use the pooled cross-sections to improve precision. This is why the paper treats the dollar-share target and the durable-balance-sheet targets slightly differently.

Moment mapping. The baseline SMM targets are: (i) the mean years since the last durable adjustment, (ii) the mean durable share of total household wealth, and (iii) the unconditional mean dollar share of liquid assets. I compute each target to mirror the model objects as closely as possible: the durable share uses housing plus vehicles over total wealth, and the dollar share uses foreign-currency deposits over total liquid assets. Since EFHU is cross-sectional, I compare these moments to the model’s stationary distribution rather than to panel transitions.

Survey design and inference. I compute the descriptive statistics and empirical moments using household survey weights. I construct standard errors and the SMM weighting matrix using

a household-level block bootstrap, resampling households with replacement to preserve cross-variable dependence within households.

B ENGIH 2016: Expenditure Moments and Non-Targeted Validation

The appendix describes how I construct moments from the 2016 Encuesta de Gastos e Ingresos de los Hogares (ENGIH 2016) and report their values. The moments are not targeted in the SMM estimation but serve two purposes: (i) they discipline the calibration of v and (ii) they provide an external check on the model's predicted allocation between durable and non-durable expenditure and on household savings behavior.

Data and measurement. ENGIH 2016 is a nationally representative household expenditure survey conducted by the Instituto Nacional de Estadística (INE), covering 6,889 Uruguayan households. I use transaction-level expenditure records classified by detailed COICOP item and by broad expenditure division, together with the survey's monthly-equivalent spending measure. I compute all moments using the official household expansion weights. Household income is measured as total household income including imputed rent, expressed in monthly Uruguayan pesos.

Durable and non-durable classification. I classify expenditure into durables and non-durables by COICOP division:

- *Non-durables*: divisions 01 (food), 02 (beverages/tobacco), 03 (clothing), 04 (housing and utilities), 06 (health), 08 (communications), 09 (recreation), 10 (education), 11 (restaurants), 12 (miscellaneous).
- *Durables*: division 05 (furnishings and household equipment) and division 07 (transport, including vehicle purchases and associated costs).
- Divisions 13–17 (taxes, financial charges, and imputed transfers) are excluded from the consumption denominator as they are not consumption expenditure.

Monthly-equivalent values are used throughout to ensure comparability across items with different reference periods.

Moments. Table 8 reports the main moments computed from ENGIH 2016. The non-durable budget share (88.5% mean, 91.3% median) directly supports the calibration $\nu = 0.88$. The durable investment rate and savings moments are not targeted by the SMM estimator but serve as informal checks on the model’s predictions.

Table 8: ENGIH 2016 expenditure moments (non-targeted)

Moment	Mean	Median
<i>Budget shares (share of consumption expenditure, DIV 01–12)</i>		
Non-durable budget share	0.885	0.913
Durable budget share (DIV 05+07)	0.116	0.087
<i>Durable investment (monthly-equivalent expenditure / income)</i>		
Durable investment rate (DIV 05+07)	0.121	0.073
Share of HHs with positive durable expenditure	97.5%	
Vehicle purchase rate (reference period)	7.3%	
<i>Savings (income – non-durable consumption) / income</i>		
Household savings rate	0.066	0.161
Share with positive savings	69.1%	
10th-percentile savings rate	–0.356	
90th-percentile savings rate	0.459	
<i>Asset ownership</i>		
Vehicle ownership rate	46.2%	
Homeownership rate	59.2%	

Notes: All moments are survey-weighted using household expansion weights. Consumption expenditure is defined as the sum of monthly-equivalent expenditures across COICOP divisions 01–12. Income is total household income including imputed rent. The savings rate is household income minus non-durable expenditure, divided by household income; negative values correspond to households whose non-durable expenditure exceeds reported income. Vehicle ownership is constructed from the household roster’s vehicle-holding indicator. Homeownership is constructed from the survey’s housing-tenure classification, with owner-occupied categories grouped together. N = 6,889 households.

Interpretation. The model’s non-durable utility share $\nu = 0.88$ matches the survey mean budget share of 88.5% closely. The durable investment rate of 12.1% (mean) and 7.3% (median) of household income reflects both routine durable maintenance included in division 07 and lumpy vehicle and appliance purchases. The wide dispersion of the savings rate (10th-percentile = –35.6%, 90th-percentile = +45.9%) is consistent with the model’s prediction that households with large durable adjustment episodes show negative savings rates in the adjustment period, while non-adjusting households accumulate liquid wealth. Vehicle ownership of 46.2% provides a cross-sectional benchmark for the ergodic share of durable holders in the simulated economy.

C Computational Procedure

Discretization. I solve the model on a finite grid for the state variables (w, d, e, y) . I approximate idiosyncratic income and exchange-rate processes by n_y - and n_e -state Markov chains using Tauchen’s method. Asset and durable grids are nonlinearly spaced to concentrate points near borrowing constraints and low durable holdings; the baseline grid sizes are $n_w = 15$ and $n_d = 21$. Durables depreciate deterministically at rate δ .

Solution method. For each state-grid point, I solve the Bellman equations associated with the adjustment and no-adjustment branches, $V^{\text{adjust}}(S)$ and $V^{\text{noadjust}}(S)$. I handle off-grid choices by interpolation, and I take expectations over the discrete transition matrices for (e', y') . Value function iteration proceeds until convergence, $\|V^{(k+1)} - V^{(k)}\|_\infty < 10^{-4}$. I store optimal policy functions for consumption, portfolio choice, and durable adjustment for simulation.

Simulation. Given parameter vector θ , I simulate the model forward for $T = 1,500$ periods, discarding the first 250 as burn-in. I compute moments from the ergodic distribution of simulated households and align them with their empirical counterparts (means, dispersions, ratios, and adjustment frequencies). I use common random numbers across parameter evaluations to reduce Monte Carlo noise in the SMM objective.

Estimation routine. Estimation uses a multistart random global search over the parameter bounds. Within each start, I draw candidate parameter vectors uniformly from the admissible region and retain the best candidate found. I then repeat the search across many starts with different random seeds and initial draws, which yields tens of thousands of objective-function evaluations overall. The approach is robust to the non-smooth objective surface induced by discrete durable adjustment, which makes gradient-based methods unreliable. The best candidate found is always preserved across search restarts.

Weighting matrix and inference. I construct the optimal SMM weighting matrix W^* from the empirical covariance of moments, estimated via influence functions and a block bootstrap clustered by survey wave. After re-estimation under W^* , I quantify parameter uncertainty using the delta method as described in Appendix D.

Diagnostics and robustness. I verify that simulated ergodic distributions replicate the targeted data moments and report the over-identifying J -statistic. All results are robust to alternative random seeds, grid densities, and simulation horizons. Sensitivity checks for key parameters confirm policy functions and moment matches are locally stable.

General-equilibrium extension. To assess GE robustness, I supplement the PE baseline with an Aiyagari-style bisection over the domestic peso interest rate r . Given a candidate r , the representative firm with Cobb-Douglas technology $Y = ZK^\alpha$ implies capital demand $K(r) = [\alpha Z / (r + \delta_K)]^{1/(1-\alpha)}$ and wage $w(r) = (1 - \alpha)ZK(r)^\alpha$. The VFI and simulation are re-solved at $(r, w(r))$, holding all household parameters fixed at their estimated values, to obtain aggregate peso deposits $A(r)$. The algorithm bisects until $|A(r) - K(r)| < \varepsilon$. TFP Z is calibrated so that $w(r_{\text{PE}}) = 1$, ensuring the GE bisection operates in the same price space as the PE baseline. Dollar savings and durable purchases clear in the world market and do not enter the domestic capital-market condition.

D Estimation

Objective. I estimate parameters by Simulated Method of Moments (SMM). Let m^{data} denote the vector of empirical moments and $m(\theta)$ the corresponding simulated moments generated by the model under parameter vector θ . The estimator solves

$$\hat{\theta} = \arg \min_{\theta} Q(\theta), \quad Q(\theta) = [m(\theta) - m^{\text{data}}]^\top W [m(\theta) - m^{\text{data}}].$$

I initialize the procedure with a diagonal weighting matrix

$$W_0 = \text{diag} \left(1 / \widehat{\text{Var}}(m_k^{\text{data}}) \right).$$

After an initial pass, I estimate the covariance matrix of the empirical moments using influence functions and a block bootstrap clustered by survey wave. The efficient weighting matrix is then $W^* = \widehat{\Sigma}^{-1}$, where $\widehat{\Sigma}$ denotes the estimated covariance of m^{data} . I re-estimate parameters using W^* and report the over-identifying J -statistic and its associated p -value.

Simulation and optimization. I compute simulated expectations by forward simulation under the estimated Markov processes for income and exchange rates. I use common random numbers across parameter evaluations to reduce simulation noise and ensure smoothness of the objective

function.

I perform optimization via a multistart random global search over the parameter bounds. Within each start, I draw candidate parameter vectors uniformly from the admissible region, evaluate $Q(\theta)$ repeatedly, and retain the best candidate found. I then repeat the search across many starts with different random seeds and initial draws, which yields tens of thousands of objective-function evaluations overall. I save the incumbent best candidate throughout the search. The procedure is computationally intensive but avoids reliance on gradient information, which is unreliable given the non-smooth objective induced by discrete durable adjustment.

Inference and robustness. I quantify parameter uncertainty using the delta method. Let $G = \partial m(\theta)/\partial \theta^\top|_{\hat{\theta}}$ denote the Jacobian of simulated moments evaluated at the estimate. The asymptotic covariance matrix is

$$\widehat{\text{Var}}(\hat{\theta}) = (G^\top W^* G)^{-1} G^\top W^* \hat{\Sigma} W^* G (G^\top W^* G)^{-1}.$$

I report standard errors as square roots of the diagonal elements.

I assess numerical robustness by varying simulation horizons, random seeds, and initial conditions, and by reporting sensitivity profiles for key structural parameters (f, κ) .

D.1 Augmented five-moment re-estimation

As an additional robustness check, I re-estimate the model using five moments rather than three. The augmented target vector keeps the baseline moments (mean adjustment duration, mean durable-wealth share, and mean dollar share of liquid assets) and adds two portfolio moments: the volatility of the dollar share and dollar participation.

The augmented run yields $(\hat{f}, \hat{\kappa}) = (0.02254, 0.02417)$ with objective value $Q = 0.734$. Relative to the baseline interpretation, the key point is stability of the structural mechanism rather than a large parameter shift.

Table 9 reports the augmented-fit diagnostics.

Table 9: Appendix Table D1: Augmented five-moment fit diagnostics

Moment	Data	Data SE	Model
Duration mean	23.178	0.214	28.364
Durable-wealth mean	0.829	0.010	0.472
Dollar share mean	0.0796	0.004	0.148
Dollar-share volatility	0.0630	0.0035	0.101
Dollar participation	0.386 [†]	0.0064	0.246

Notes: Data standard errors for the first three moments are the same bootstrap SEs reported in Table 4. For dollar-share volatility and dollar participation, SEs are block-bootstrap estimates (500 replications). [†]The dollar participation data moment (0.386) uses the pooled three-wave EFHU sample. Waves 1 and 3 record currency composition in discrete ranges that cannot register a zero dollar share, mechanically coding all respondents as participants; the pooled rate therefore exceeds the Wave-2 continuous measure (0.106). The augmented run uses the pooled moment for comparability across waves but the gap relative to the Wave-2 figure should be borne in mind.

These results support the same qualitative reading as the baseline: the model captures the direction of the core mechanisms, while the dollarization block remains the less tightly disciplined dimension of the specification.

E Durable adjustment with dollar pricing: gaps, inaction, and hedging

The appendix provides intuition for the household’s durable adjustment decision in the presence of dollar pricing and exchange-rate risk. The discussion is illustrative and focuses on the reduced-form environment used in the quantitative model.

Environment. The household state is

$$S_t = (w_t, d_{t-1}, e_t, y_t),$$

where w_t is total liquid wealth in local-currency units, d_{t-1} is the beginning-of-period durable stock, e_t is the nominal exchange rate (local currency per USD), and y_t is an idiosyncratic income shifter. Preferences are

$$u(c_t, d_t) = \frac{(c_t^\nu d_t^{1-\nu})^{1-\gamma}}{1-\gamma}, \quad \nu \in (0, 1), \gamma > 0.$$

Durables are priced in dollars at price p_d , so the local replacement cost is $P_t^d = e_t p_d$.

Durable adjustment is lumpy due to partial irreversibility. If the household does not adjust, the durable stock evolves as

$$d_t = (1 - \delta)d_{t-1}.$$

If the household adjusts, it chooses a new durable stock d_t and liquidates the depreciated stock at a resale value subject to a haircut $f \in (0, 1)$.

Frictionless target and normalized gap

Absent adjustment frictions, the optimal durable choice satisfies the intratemporal condition

$$\frac{u_d(c_t, d_t)}{u_c(c_t, d_t)} = P_t^d,$$

which implies the frictionless target

$$d_t^* = \frac{1 - v}{v} \cdot \frac{c_t}{e_t p_d}.$$

Define the normalized durable gap as

$$z_t \equiv \frac{(1 - \delta)d_{t-1}}{d_t^*} = \underbrace{\frac{(1 - \delta)v}{1 - v}}_{\xi} \cdot \frac{e_t p_d d_{t-1}}{c_t}.$$

Values $z_t > 1$ correspond to an excessive durable stock relative to the target, while $z_t < 1$ indicates underaccumulation.

Exchange-rate movements directly affect the gap: a depreciation raises $e_t p_d$, increasing z_t and pushing households toward delaying replacement or downsizing.

Inaction and asymmetric adjustment

Since adjusting involves resale losses, households tolerate deviations from the frictionless target. The value gain from adjustment is decreasing in $|z_t - 1|$ near the target and increasing when the gap becomes large. The friction generates an inaction region in z_t .

Unlike fixed-cost S-s models, partial irreversibility implies asymmetric inaction:

- When $z_t > 1$ (excess durable stock), selling is costly due to the resale haircut, so households tolerate relatively large deviations.
- When $z_t < 1$ (insufficient durable stock), buying is less distorted, leading to a narrower inaction region.

As a result, the effective inaction set is wider on the “sell” side than on the “buy” side, consistent with infrequent durable replacement and long observed tenures.

Dollar saving and hedging

Households allocate end-of-period savings across peso and dollar assets. Let s_t denote the dollar share of savings. The next-period liquid wealth is

$$w_{t+1} = (1+r)(1-s_t)w'_t + (1+r^{\$})s_t w'_t \frac{e_{t+1}}{e_t} - \kappa s_t w'_t,$$

where κ captures the reduced liquidity of dollar saving.

Dollar assets hedge exchange-rate risk: when e_{t+1} rises, the local-currency value of dollar savings increases, partially offsetting the higher replacement cost of durables. By stabilizing future consumption relative to durable needs, dollar saving reduces the volatility of the marginal rate of substitution between c and d .

Remark E.1 (Hedging and inaction). Policies or environments that increase the effectiveness of dollar saving as a hedge against exchange-rate risk reduce the volatility of the normalized gap z_t and shrink the effective inaction region.

Analytical comparative statics

Locally, the normalized gap satisfies

$$z_t \propto \frac{e_t p_d d_{t-1}}{c_t}.$$

Hence:

$$\frac{\partial z_t}{\partial e_t} > 0 \quad \text{and} \quad \frac{\partial z_t}{\partial y_t} < 0.$$

Depreciations push households toward inaction or downsizing, while positive income shocks encourage upgrading. Increases in irreversibility (f) widen the inaction region, whereas improved hedging opportunities through dollar saving narrow it.

The mechanism clarifies how dollar pricing, exchange-rate risk, and portfolio choice jointly shape durable adjustment behavior in the quantitative model.

Figure 8 connects the analytical gap object to the simulated stationary distribution. The figure reports only the stationary density of the durable gap, focusing on the economically relevant range from -5 to 5 log points. Most mass lies close to zero, where households remain near their target durable stock and adjustment is infrequent. The thin tails indicate that large misalignments are uncommon in the stationary distribution, consistent with a model in which households eventually pay the nonconvex adjustment cost when the gap becomes too large.

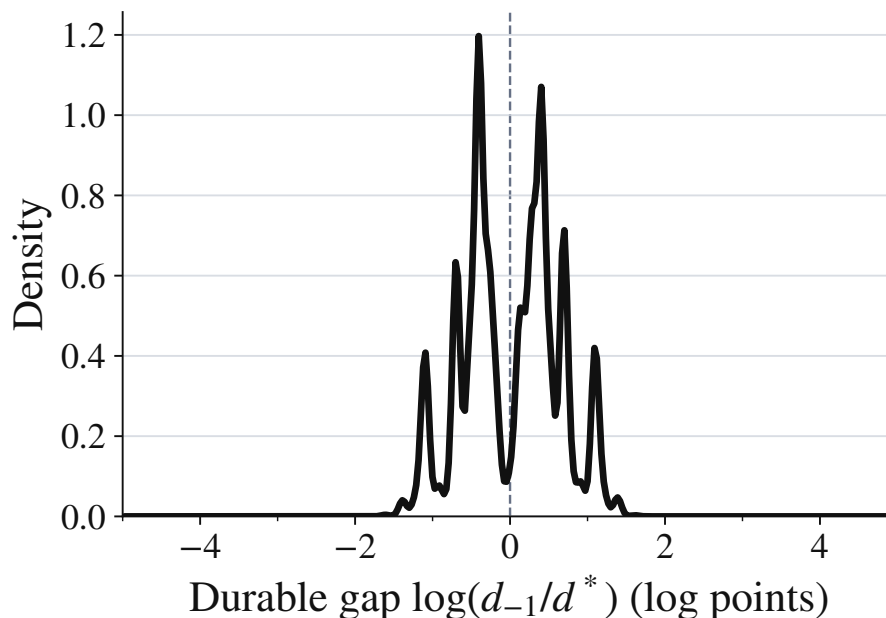


Figure 8: Stationary density of the durable gap

Notes: The figure plots the stationary density of the durable gap over the interval $[-5, 5]$ log points. The concentration of mass near zero indicates that most households remain relatively close to their target durable stock in the stationary distribution.

F Convex Durable Adjustment Model

The section presents a convex-adjustment benchmark that clarifies the interaction between exchange-rate risk, dollar assets, and durable investment. The environment abstracts from nonconvex adjustment costs and partial irreversibility, allowing for smooth durable adjustment. The purpose is interpretive: to isolate how dollar pricing mechanically links portfolio choice and durable investment through Euler equations.

Environment. Households choose consumption c_t , liquid wealth, dollar assets, and durable investment $x_t \equiv d_t - (1 - \delta)d_{t-1}$. Durables are priced in dollars at price p_d , so the local-currency purchase price is $e_t p_d$. Adjustment costs are quadratic:

$$\frac{\phi}{2} x_t^2.$$

Let λ_t denote the multiplier on the budget constraint and μ_t the multiplier on durable accumulation.

(i) USD asset Euler. The optimality condition for dollar-denominated liquid assets is

$$1 = \mathbb{E}_t \left[\beta (1 + r^{\$}) \frac{\lambda_{t+1}}{\lambda_t} \frac{e_{t+1}}{e_t} \right].$$

Dollar assets pay off in local currency through exchange-rate revaluation. Hence, exchange-rate risk enters directly into the intertemporal trade-off.

(ii) Durable Euler (Tobin's q). Define Tobin's q_t as the shadow value of installed durables relative to their replacement cost:

$$q_t \equiv \frac{\mu_t}{\lambda_t e_t p_d}.$$

With quadratic adjustment costs, the investment rule is linear:

$$q_t = 1 + \phi x_t.$$

The forward-looking Euler equation for q_t is

$$q_t = \frac{u_d(c_t, d_t)}{\lambda_t e_t p_d} + \mathbb{E}_t \left[\beta (1 - \delta) \frac{\lambda_{t+1}}{\lambda_t} \frac{e_{t+1}}{e_t} q_{t+1} \right].$$

(iii) Intratemporal MRS. Preferences imply

$$\frac{u_d}{u_c} = \frac{1 - \nu}{\nu} \frac{c_t}{d_t}.$$

Interpretation. Tobin's q_t measures the marginal value of an additional unit of durable relative to its dollar replacement cost. Exchange-rate risk enters both Euler equations symmetrically through the term e_{t+1}/e_t :

- In the *portfolio Euler*, dollar assets hedge depreciations by delivering higher local-currency

payoffs.

- In the *durable Euler*, depreciations raise the future replacement cost, increasing the continuation value of installed capital.

As a result, exchange-rate movements jointly affect saving in dollar assets and the incentive to invest in durables. Dollar assets hedge precisely the states in which durable replacement becomes expensive. The convex benchmark makes clear the interaction between currency choice and durable investment arises even absent nonconvexities.

In the full model, nonconvex adjustment costs and irreversibility transform smooth responses into infrequent, state-dependent adjustment episodes, but the underlying hedging logic remains the same.

G General-Equilibrium Robustness

Uruguay is a small open economy: the exchange rate and world interest rates are externally determined, so the partial-equilibrium framework is the natural baseline. This appendix section provides a general-equilibrium robustness check following Aiyagari (1994) that endogenizes domestic factor prices through the firm block, while holding all household-side parameters fixed at their estimated values. The exercise asks whether the welfare results change materially once the domestic savings market clears in equilibrium rather than taking prices as fixed.

The exercise is motivated in part by the logic of Berger and Vavra (2015), Khan and Thomas (2008), and Bachmann et al. (2013), who show that with nonconvex adjustment and multiple accumulation margins, lumpy microeconomic reallocation need not translate into proportionally volatile aggregate quantities in general equilibrium. In their closed-economy framework the resource constraint is $Y = C + I_k + I_d$: when lumpy durable adjustment generates departures of I_d from a frictionless benchmark, the liquid capital margin I_k absorbs these changes without requiring C to be equally volatile. In the present SOE setting, durables are dollar-priced imports, so durable purchases are financed from dollar savings and clear in the world goods market; the domestic resource constraint simplifies to $Y = C + I_k$. The same buffering logic still applies: households hold both peso deposits a and durable wealth d ; when an exchange-rate movement induces some households to adjust durables, the liquid peso savings margin absorbs the reallocation without requiring amplified consumption responses.

Definition G.1 (Stationary General Equilibrium). The economy consists of a continuum of households and a representative domestic firm with Cobb-Douglas technology $Y = ZK^\alpha$, $\alpha \in (0, 1)$, where K is the aggregate capital stock, labor is inelastic and normalized to one, and Z is fixed at its stationary level. The foreign interest rate $r^\$$ and the exchange rate e remain exogenous. A *stationary general equilibrium* is a tuple $(r^*, w^*, K^*, \mu^*, \{c^*, w'^*, s^*, d^*\})$ consisting of equilibrium prices, a capital stock, a probability measure μ^* over the state space $\mathcal{S} = \mathbb{R}_+ \times \mathbb{R}_+ \times \mathcal{E} \times \mathcal{Y}$, and household decision rules, such that:

- (i) **Household optimality.** Given r^* , the decision rules $\{c^*(S), w'^*(S), s^*(S), d^*(S)\}$ solve the household problem (3.7)–(3.12) for each state $S \in \mathcal{S}$.
- (ii) **Stationarity.** μ^* is the stationary distribution induced by the decision rules and the exogenous Markov processes for (e, y) : for every measurable set $B \subseteq \mathcal{S}$,

$$\mu^*(B) = \int_{\mathcal{S}} \mathbf{1}[S' \in B] Q(S, dS') d\mu^*(S),$$

where $Q(S, \cdot)$ is the transition kernel implied by the policy functions and the exogenous processes.

- (iii) **Firm optimality.** The representative firm maximizes profits. With inelastic labor $L = 1$, equilibrium prices satisfy:

$$r^* + \delta_K = \alpha Z (K^*)^{\alpha-1}, \quad w^* = (1 - \alpha) Z (K^*)^\alpha.$$

- (iv) **Asset market clearing.** Aggregate peso deposits equal the equilibrium capital stock:

$$\int a^*(S) d\mu^*(S) = K^*,$$

where $a^*(S) = (1 - s^*(S)) w'^*(S)$ are peso deposits under the optimal portfolio rule. Dollar deposits $s^*(S) w'^*(S)/e$ are net foreign assets and clear in the world capital market; they do not enter the domestic capital stock. Durable goods are imported and financed from dollar savings; no separate domestic durable goods market clearing is required.

I solve for r^* as follows. First, I calibrate the firm's TFP level Z so that the firm block reproduces the PE wage $w_{PE} = 1$ at the PE interest rate r_{PE} :

$$Z = \left[\frac{w_{PE}}{(1 - \alpha) (\alpha / (r_{PE} + \delta_K))^{\alpha/(1-\alpha)}} \right]^{1-\alpha}.$$

This normalization ensures the GE bisection operates in the same price space as the PE calibration. I then bisect over $r \in [r_{\min}, r_{\max}]$: at each candidate r , (i) map r into implied firm capital demand $K(r)$; (ii) compute the implied wage $w(r)$ from the same firm block; (iii) solve the household VFI at those prices; (iv) simulate the stationary distribution; (v) compute aggregate peso savings $A(r, w)$; and (vi) update the bracket from the sign of $A(r, w) - K(r)$. The algorithm converges when $|A(r, w) - K(r)| < \varepsilon$. All household parameters are held fixed at $(\hat{f}, \hat{\kappa})$; no re-estimation is performed.

I report two sets of outcomes: (i) equilibrium prices and capital (r^*, w^*, K^*) together with market-clearing diagnostics; and (ii) welfare comparisons for the main stationary counterfactuals recomputed at GE prices.

Full GE results

The Aiyagari-style bisection converges to $r^* = 4.20\%$ after 42 iterations, with market-clearing residual $|A^* - K^*| = 0.940$. The bisection bracket collapses to 10^{-6} precision in r ; the residual reflects a grid-induced discontinuity in the excess demand function near the equilibrium rate and is within the tolerance set for this robustness exercise. Table 11 collects the full results.

GE impulse response diagnostics

To verify that the short-run transmission mechanism is preserved at GE prices, Table 10 compares the generalized impulse responses to a one-standard-deviation depreciation shock, computed under PE and GE stationary prices with the same shock process. The main finding is that the on-impact consumption response is almost entirely absorbed in GE (-0.86% in PE versus ≈ 0 in GE), while the durable and savings responses remain of similar sign and order. The dampening of the consumption channel at GE prices is consistent with the mechanism in the body text: the higher peso return r^* lowers the marginal cost of saving, cushioning the consumption drop, while the lower wage w^* simultaneously reduces income exposure.

Table 10: GE vs. PE generalized impulse responses (one-s.d. depreciation shock)

Variable	Impact response		Peak response (abs. value)	
	PE	GE	PE	GE
Consumption c	-0.861	≈ 0	0.861 ($t=0$)	0.306 ($t=1$)
Durable stock d	-0.188	-0.250	0.188 ($t=0$)	0.348 ($t=1$)
Liquid wealth w	+0.284	+0.106	0.284 ($t=0$)	0.123 ($t=1$)
Dollar share s	-0.009	-0.005	0.009 ($t=0$)	0.005 ($t=0$)

Notes: Responses are percentage deviations from the respective stationary distribution (PE or GE). Same shock process ($\sigma_e = 0.063$, $\rho_e = 0.85$) and household parameters ($\hat{f}$, $\hat{\kappa}$) used in both columns; only the baseline prices differ. The near-zero consumption impact in GE reflects the cushioning effect of the higher peso return on precautionary saving.

Table 11: General-equilibrium robustness results

Object	Partial equilibrium	General equilibrium
<i>A. Equilibrium prices and market clearing</i>		
Domestic peso interest rate r	0.0142	0.0420
Wage w	1.000	0.740
Capital demand K	14.35	6.22
Aggregate peso deposits A	59.06	5.28
Market-clearing residual $ A - K $	n/a	0.940
Solver convergence	n/a	Converged (bisection, 42 iterations)
<i>B. Welfare counterfactuals (CEV, %)</i>		
Higher adjustment cost ($f = 0.043$)	-20.99	-0.54
Peso-priced durables	-10.66	-0.22
Restricted dollar saving ($\kappa = 0.05$)	-9.58	-0.06

Notes: Panel A reports equilibrium prices and market-clearing diagnostics. The PE “aggregate peso deposits” entry is the simulated A at PE prices ($r_{PE} = 1.42\%$, $w_{PE} = 1.00$), which substantially exceeds domestic capital demand, reflecting the SOE’s access to world capital markets. The GE bisection clears the *domestic* peso deposit market; dollar savings clear in the world capital market and do not enter the domestic clearing condition. Panel B reports CEV welfare counterfactuals recomputed at GE prices; the near-zero GE entries indicate that equilibrium price adjustment absorbs most welfare effects visible in PE, so PE magnitudes are best read as an upper bound. See Appendix G for full solver diagnostics.

Closed-Economy Extension: I_d in the Resource Constraint

For completeness, this subsection contrasts the SOE resource constraint with the closed-economy version of Berger and Vavra (2015), where durable goods are domestically produced. The comparison clarifies that the paper’s $Y = C + I_k$ condition is not merely a simplification but follows directly from Uruguay’s import structure for durables.

In a closed economy, all durable investment is domestically absorbed, so the aggregate resource constraint reads

$$Y_t = C_t + I_{k,t} + I_{d,t},$$

where $I_{d,t} = \int p_d [d'(S) - (1 - \delta)d]^+ d\mu_t$ is aggregate gross durable investment at the domestic price p_d . Because durables are now produced domestically, the asset-market clearing must jointly satisfy

$$\int [(1 - s(S)) w'(S)] d\mu^* = K, \quad \int p_d d(S) d\mu^* = p_d D^*,$$

with p_d endogenously determined to clear the domestic durable goods market. This adds a second fixed-point dimension—simultaneous clearing in both liquid capital (K) and durable goods (D^*) markets—and requires specifying a domestic durable-producing sector.

In the present paper, Uruguay imports the vast majority of its durable goods. Durable purchases are therefore financed from dollar savings and clear in the world goods market; p_d is pinned by world prices with full pass-through. The domestic resource constraint simplifies to $Y = C + I_k$, and no domestic durable-goods market clearing is required. This is the condition imposed in Definition G.1 and in the GE bisection algorithm above. Introducing an endogenous p_d via a closed-economy fixed point would require a domestic durable-sector specification with no direct Uruguayan empirical counterpart and would substantially complicate estimation; the SOE assumption therefore reflects both institutional structure and parsimony.

H Additional Welfare Counterfactuals and Robustness

H.1 Welfare Measure Robustness

The main text reports welfare effects using the standard ex-ante consumption equivalent variation (CEV), defined as the fraction λ of baseline consumption that leaves a household indifferent

between the baseline and the counterfactual economy:

$$\lambda = \left(\frac{\mathcal{W}^{BA}}{\mathcal{W}^{AA}} \right)^{\frac{1}{1-\gamma}} - 1,$$

where $\mathcal{W}^{AA} = \sum_s v^A(s) \mu^A(s)$ is the baseline value function integrated over the baseline ergodic distribution μ^A , and $\mathcal{W}^{BA} = \sum_s v^B(s) \mu^A(s)$ evaluates the counterfactual value function at the same baseline distribution. A positive λ indicates a welfare gain from the counterfactual.

The measure is standard in the macroeconomics literature (Lucas, 1987; Aiyagari, 1994; Heathcote et al., 2008) and captures the combined effect of changes in mean consumption levels and changes in risk exposure. To assess robustness and disentangle the two components, Table 12 reports three alternative measures.

Cross-steady-state CEV (λ^{SS}). The measure compares the two ergodic steady states directly:

$$\lambda^{SS} = \left(\frac{\mathcal{W}^{BB}}{\mathcal{W}^{AA}} \right)^{\frac{1}{1-\gamma}} - 1,$$

where $\mathcal{W}^{BB} = \sum_s v^B(s) \mu^B(s)$ uses both the counterfactual value function and the counterfactual ergodic distribution. Relative to λ , the measure includes the welfare effect of households re-optimizing their portfolios and durable holdings under the new policy environment.

Lucas variance approximation (λ^{Lucas}). Following Lucas (1987), welfare costs can be approximated by

$$\lambda^{\text{Lucas}} = -\frac{1}{2} \gamma [\text{Var}(\log c^{CF}) - \text{Var}(\log c^{\text{base}})].$$

The measure isolates the *insurance* or *variance* channel, abstracting from level effects in mean consumption. It is the most conservative of the three, as it ignores portfolio reoptimization and mean wealth effects. Positive values indicate that the counterfactual reduces consumption volatility.

Mean consumption level change ($\Delta \bar{c}$). The complement to λ^{Lucas} is the pure level effect:

$$\Delta \bar{c} = \frac{\mathbb{E}^B[c]}{\mathbb{E}^A[c]} - 1.$$

Together, λ^{Lucas} and $\Delta \bar{c}$ decompose the total welfare effect into a risk-reduction component and a level component. When the two measures disagree in sign with the primary CEV, it signals that

the level and variance effects work in opposite directions.

Interpretation of CEV estimates. Counterfactual CEVs combine mean-consumption and risk channels. The Lucas approximation λ^{Lucas} isolates the variance channel and serves as a conservative lower bound. Results in the main tables are best interpreted alongside λ^{Lucas} and $\Delta\bar{c}$.

H.2 Additional Counterfactuals

Table 12 reports welfare decompositions for counterfactuals retained in the main text, using all four measures.

Table 12: Welfare decompositions for selected counterfactuals

Counterfactual	Welfare measures				Consumption volatility	
	λ (%)	λ^{SS} (%)	λ^{Lucas} (%)	$\Delta\bar{c}$ (%)	$\sigma(\log c)$ base	$\sigma(\log c)$ CF
<i>A. Durable pricing denomination</i>						
Peso-priced durables	−10.66	−12.41	−2.50	+45.74	0.549	0.571
<i>B. Rare disaster benchmark</i>						
Barro-style rare disaster benchmark	+0.06	+0.08	+0.92	−2.58	0.549	0.540

Notes: λ is the ex-ante CEV at the baseline ergodic distribution (primary measure, (5.1)). λ^{SS} is the cross-steady-state CEV. $\lambda^{\text{Lucas}} = -\frac{1}{2}\gamma[\text{Var}(\log c^{CF}) - \text{Var}(\log c^{\text{base}})]$ isolates the variance channel only. $\Delta\bar{c} = \mathbb{E}^B[c]/\mathbb{E}^A[c] - 1$ captures the pure mean consumption level effect. Positive values indicate a welfare gain in the counterfactual economy. All experiments use estimated baseline parameters. Row A uses a modified model in which the durable purchase and sale price is denominated in pesos rather than dollars, so that the peso cost of durables does not co-vary with the exchange rate. Dollar savings still bear exchange-rate risk. Row B reports a Barro-style rare-disaster benchmark: a low-probability tail event in the spirit of Barro (2006), implemented here as a large exchange-rate depreciation.

Peso-priced durables. In the counterfactual, the peso price of durables is fixed and does not respond to exchange-rate movements; dollar savings continue to bear exchange-rate risk. The estimate $\lambda = -10.62\%$ indicates a welfare loss from removing dollar pricing in the model. The decomposition remains informative: while $\Delta\bar{c}$ is positive, consumption risk rises and the Lucas component is negative, consistent with the loss of the denomination hedge. Quantitatively, components in the decomposition should be interpreted with caution because they are sensitive to model fit and to how portfolios re-optimize across regimes.

Barro-style rare disaster benchmark. The rare-disaster row implies a near-zero but slightly positive CEV in the current run. In the model, the welfare effect remains modest because the low

arrival rate keeps the tail event from dominating stationary behavior. The decomposition indicates that the variance channel is positive while mean consumption shifts slightly down, leaving a net effect close to zero in magnitude.

Distributional incidence. The distributional incidence of both counterfactuals is a natural object of interest: for the restricted dollar saving experiment, it reveals whether dollar access disproportionately benefits wealthier households (who hold more dollar assets) or poorer households (who may rely on dollars as a precautionary buffer); for the higher adjustment cost experiment, it shows which households bear the largest burden from durable illiquidity. A careful welfare decomposition by wealth quintile requires a reliable mapping between the model’s wealth distribution and the CEV formula, which in turn depends on the model’s ability to match the empirical wealth distribution. Given the current model fit limitations discussed in Section 4, distributional welfare estimates are deferred to future work with a richer model specification.

H.3 Exchange-rate welfare accounting from GIRFs

To complement the stationary welfare comparisons, I compute event-level welfare from GIRFs using the method described in Section 4.5. The GIRF-based welfare object captures the consumption-equivalent cost of a single depreciation episode and its transition dynamics. It is distinct from the stationary CEVs, which compare ergodic distributions; the GIRF measure isolates the dynamics following a specific shock, integrated from the ergodic distribution of initial states.

The method constructs shocked and unshocked paths using common future innovations, computes the finite-horizon consumption-equivalent difference along each path, and averages across initial states. I then scale by an assumed quarterly arrival probability π_q to convert the event-level cost into an annualized flow-equivalent welfare measure.

For a one-standard-deviation depreciation ($1\sigma_e$), the event-level CEV is about 0.08%. Under $\pi_q = 0.10$ (roughly once every 2.5 years), the estimate maps into about 0.008% per quarter, or approximately 0.03% annualized. The estimate places recurrent exchange-rate risk in the same order of magnitude as standard business-cycle welfare costs (0.05–0.1%; Lucas, 1987; Imrohoroğlu, 1989) and well below the stationary welfare effects of removing dollar pricing or restricting dollar access. The ordering is intuitive: the household can partially self-insure against individual shocks through the portfolio, but cannot undo the structural consequences of removing the hedging instrument entirely.

I 2002 Crisis-Process GIRF Benchmark

This section reports a crisis-process GIRF benchmark that maps the model to the 2002 Uruguay exchange-rate episode documented by Drenik et al. (2018). The objective is not to re-estimate structural parameters, but to evaluate whether the same household mechanism identified in the baseline GIRFs scales to a large depreciation process.

Operationally, I replace the baseline shock scale with the 2002 benchmark devaluation (89% in one quarter, $\log(1.89) = 0.6366$) and compute crisis GIRFs under a high-volatility exchange-rate calibration that keeps the event inside the model grid. I then compare impact and short-run responses to the baseline GIRFs. The key result is qualitative continuity: the same margins remain active—deferred durable adjustment, liquid-buffer accumulation, and portfolio rebalancing—with larger magnitudes under the crisis process.

These crisis-process GIRFs should be interpreted as event-level diagnostics rather than stationary counterfactuals. They complement Section 4.5 by showing that the paper’s core mechanism is robust to the scale of the 2002 exchange-rate event.

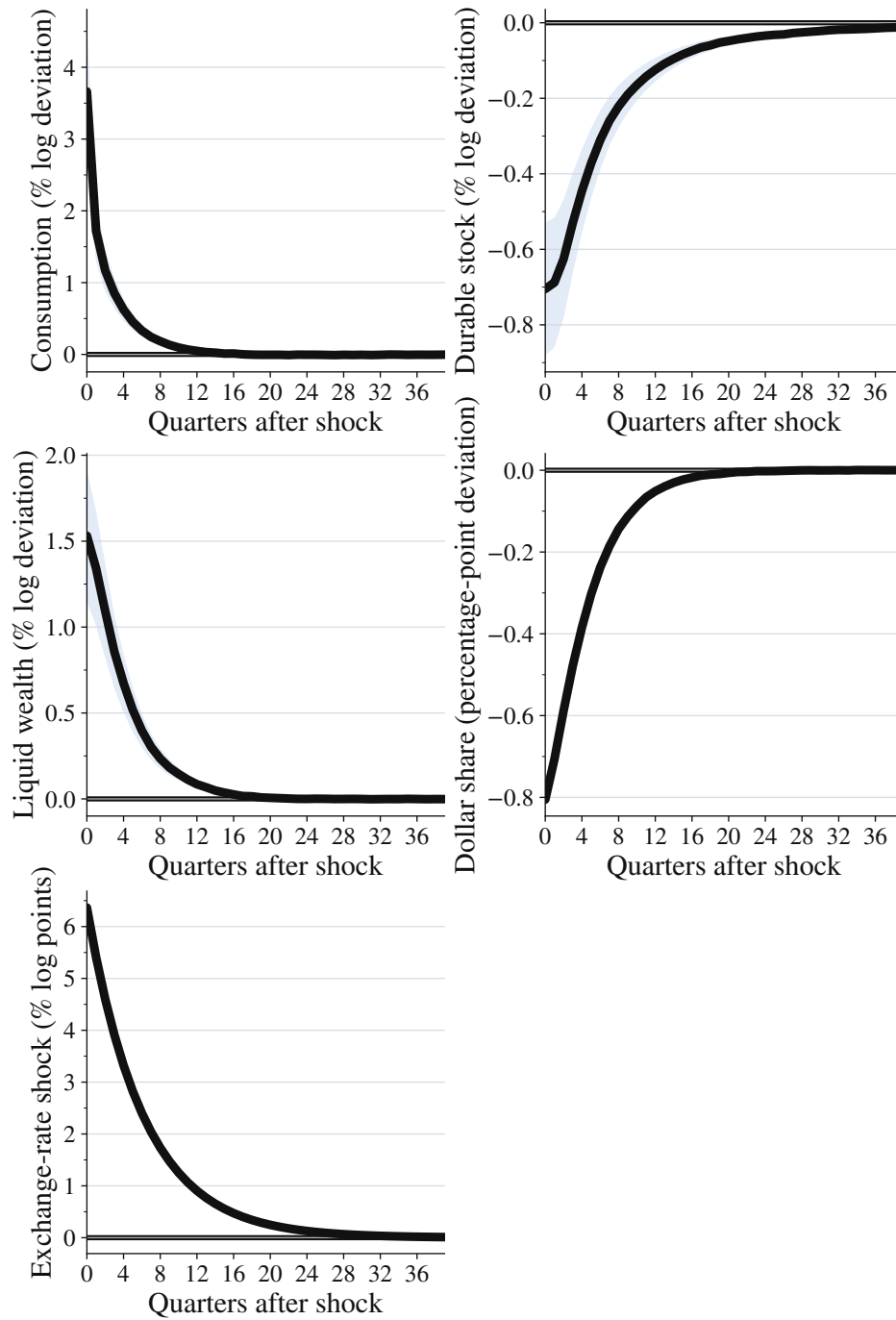


Figure 9: Crisis-process GIRF: household responses to 2002-scale devaluation (40-quarter horizon)

Notes: Generalized impulse responses following a shock calibrated to the 2002 Uruguay devaluation (89% in one quarter). Each panel shows the percentage-point difference between the shocked and baseline path, averaged over 1,000 initial-state draws from the ergodic distribution (Koop-Pesaran-Potter 1996 procedure). Shaded bands are ± 1 standard error across histories. The shock scale exceeds the baseline $1\sigma_e$ experiment; the qualitative pattern—deferred durable adjustment, liquid-buffer accumulation, portfolio rebalancing—is the same but with amplified magnitudes.

J Comparative statics

The following figures vary each structural parameter one at a time, holding all others fixed at their estimated values, and trace the response of all five moments: mean durable duration, mean and variance of the durable wealth share, mean dollar share, and dollar-share volatility. The exercise provides a visual check on identification and on the model's internal consistency.

Resale haircut f . Higher f widens the inaction region: households tolerate larger deviations from their target before paying the adjustment cost, raising duration and increasing durable wealth concentration. The dollar share rises in tandem, reflecting the stronger precautionary motive to hold dollar assets when replacement becomes more costly. The complementarity between f and the dollar share underpins the identification of the resale haircut in the SMM.

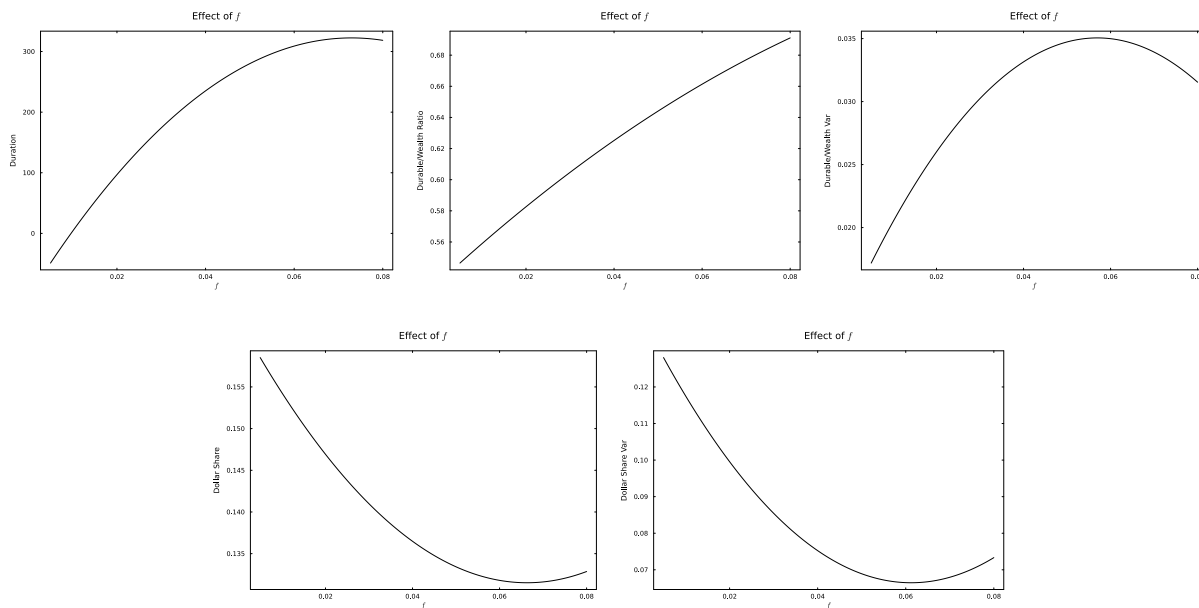


Figure 10: Comparative statics: resale haircut f

Portfolio wedge κ . Higher κ raises the cost of holding dollar assets, directly reducing the equilibrium dollar share and its cross-sectional dispersion. Effects on durable moments are indirect: by weakening the hedging channel, a higher wedge alters the timing and size of durable adjustments, shifting duration and the durable wealth distribution.

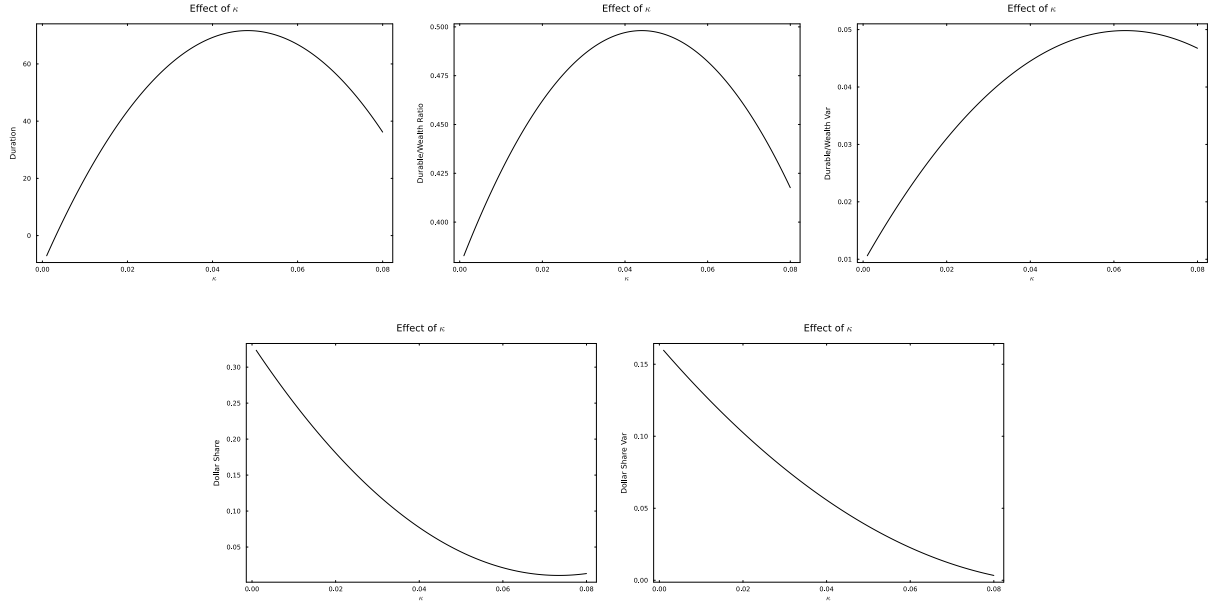


Figure 11: Comparative statics: portfolio wedge κ

Exchange-rate volatility σ_e . Greater exchange-rate uncertainty strengthens the hedging motive, raising the equilibrium dollar share and its volatility. Durable moments respond through the pass-through channel: higher σ_e increases the variance of the local-currency replacement cost, widening the inaction region and lengthening duration.

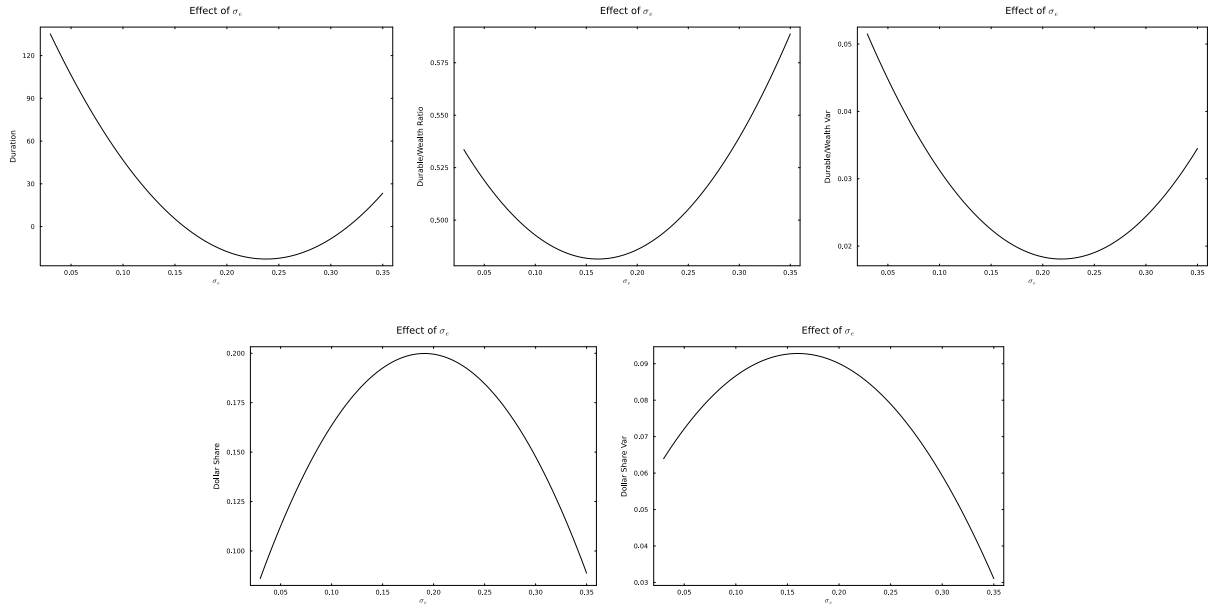


Figure 12: Comparative statics: exchange-rate volatility σ_e

Depreciation rate δ . Faster depreciation mechanically shortens durable tenure and raises the durable investment rate by compressing the effective life of each durable purchase. Portfolio moments respond indirectly: a higher δ increases the frequency of adjustment episodes, which in turn shifts the demand for dollar assets as a hedge against replacement costs.

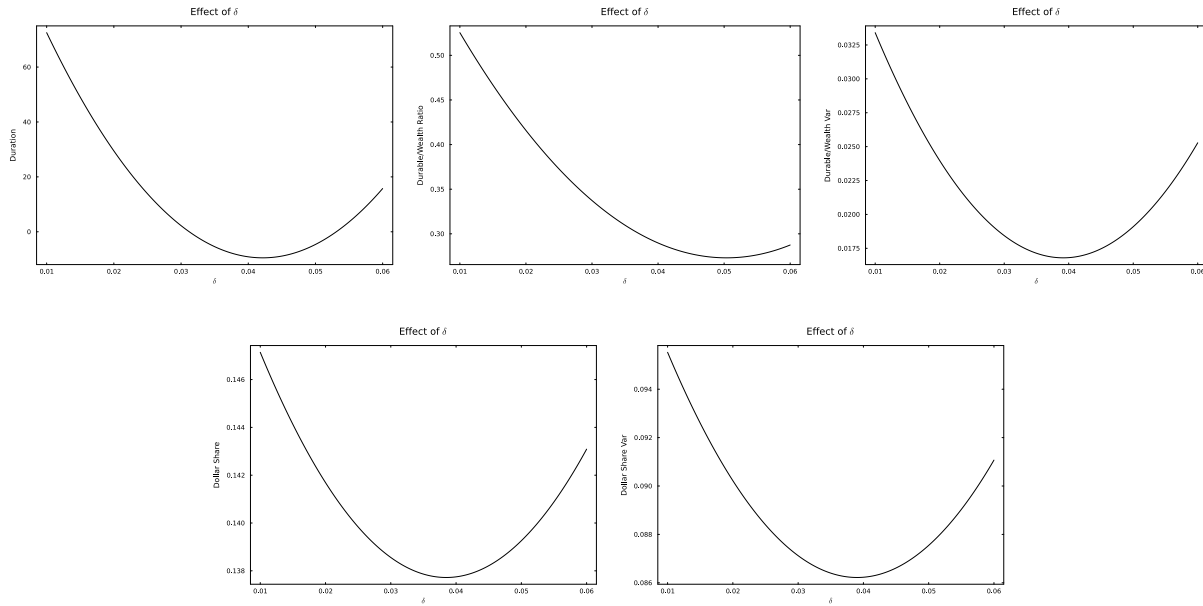


Figure 13: Comparative statics: depreciation rate δ